Algebraic Geometry 2

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1 Spectra and its Structure Sheaf

1.1 The Zariski Topology and Hilbert's Nullstellensatz for Spectra

Recall that for A a commutative ring, we defined

$$\text{Spec}(A) = \{P \subseteq A \mid P \text{ is a prime ideal of } A\}$$

We can similarly define the notion of vanishing loci of Spec(A).

Definition 1.1.1 (Zero Locus) Let R be a commutative ring. Let $T \subseteq R$. Define the vanishing locus of T to be

$$\mathbb{V}^{\mathsf{S}}(T) = \{p \in \text{Spec}(R) \mid T \subseteq p\}$$

Lemma 1.1.2 Let R be a commutative ring. Then the following are true.

- If $T \subseteq R$ is a subset and $I = (T)$ is the ideal generated by T, then
$$\mathbb{V}^{\mathsf{S}}(T) = \mathbb{V}^{\mathsf{S}}(I)$$
- If $I \subseteq R$ is an ideal of R, then
$$\mathbb{V}^{\mathsf{S}}(\sqrt{I}) = \mathbb{V}^{\mathsf{S}}(I)$$

Proof Clearly since $T \subseteq I$, we have $\mathbb{V}^{\mathsf{S}}(I) \subseteq \mathbb{V}^{\mathsf{S}}(T)$. Conversely, suppose that $P \in \mathbb{V}^{\mathsf{S}}(T)$. Then $T \subseteq P$. But $I = (T)$ is the smallest ideal containing T. Hence $(T) \subseteq P$.

Let $P \in \mathbb{V}^{\mathsf{S}}(\sqrt{I})$. Then $I \subseteq \sqrt{I} \subseteq P$ and so $P \in \mathbb{V}^{\mathsf{S}}(I)$. Conversely, suppose that $P \in \mathbb{V}^{\mathsf{S}}(I)$. Then $I \subseteq P$. Since $\sqrt{I}$ is the intersection of all prime ideals containing I, we must have $\sqrt{I} \subseteq P$. Hence $P \in \mathbb{V}^{\mathsf{S}}(\sqrt{I})$. ∎

The proposition shows that we only need to concern ourselves with the zero set of radical ideals of A.

Lemma 1.1.3

Let R be a commutative ring. Let I, J be ideals of R. Then the following are true.

- If $I \subseteq J$ are ideals of R, then $\mathbb{V}^{\mathsf{S}}(J) \subseteq \mathbb{V}^{\mathsf{S}}(I)$.
- We have $\mathbb{V}^{\mathsf{S}}(IJ) = \mathbb{V}^{\mathsf{S}}(I \cap J) = \mathbb{V}^{\mathsf{S}}(I) \cup \mathbb{V}^{\mathsf{S}}(J)$.

Proof

Let $P \in \mathbb{V}^{\mathsf{S}}(I_2)$. Then $I_2 \subseteq P$. Hence $I_1 \subseteq P$ and $P \in \mathbb{V}^{\mathsf{S}}(I_1)$.

From Commutative Algebra 1, we have

$$\mathbb{V}^{\mathsf{S}}(IJ) = \mathbb{V}^{\mathsf{S}}(\sqrt{IJ}) = \mathbb{V}^{\mathsf{S}}(\sqrt{I \cap J}) = \mathbb{V}^{\mathsf{S}}(I \cap J)$$

∎

Lemma 1.1.4 Let A be a commutative ring. The following are true.

- $\mathbb{V}^{\mathsf{S}}(1) = \emptyset$
- $\mathbb{V}^{\mathsf{S}}(0) = \text{Spec}(A)$
- Let $\{a_i \mid i \in I\}$ be a set of ideals of A, then
$$\mathbb{V}^{\mathsf{S}}\left(\sum_{i \in I} a_i\right) = \bigcap_{i \in I} \mathbb{V}^{\mathsf{S}}(a_i)$$
- Let $\{a_1, \dots, a_n\}$ be a finite set of ideals of A, then
$$\mathbb{V}^{\mathsf{S}}\left(\bigcap_{k=1}^{n} a_k\right) = \bigcup_{k=1}^{n} \mathbb{V}^{\mathsf{S}}(a_k)$$

Proof The first two are clear. We have that

$$\begin{aligned} P \in \mathbb{V}^{\mathrm{S}}\left(\sum_{i\in I} a_i\right) &\iff \sum_{i\in I} a_i \subseteq P \\ &\iff a_i \subseteq P \text{ for all } i \in I \\ &\iff P \in \mathbb{V}^{\mathrm{S}}(a_i) \text{ for all } i \in I \\ &\iff P \in \bigcap_{i\in I} \mathbb{V}^{\mathrm{S}}(a_i) \end{aligned}$$

It suffices to complete the case $n = 2$ and by induction it is true for any finite number of ideals. We have that

$$\begin{aligned} P \in \mathbb{V}^{\mathrm{S}}(a_1 \cap a_2) &\iff a_1 \cap a_2 \subseteq P \\ &\iff a_1 a_2 \subseteq a_1 \cap a_2 \subseteq P \\ &\iff a_1 \subseteq P \text{ or } a_2 \subseteq P \\ &\iff P \in \mathbb{V}^{\mathrm{S}}(a_1) \cup \mathbb{V}^{\mathrm{S}}(a_2) \end{aligned}$$

■

It is easy to see that lemma 1.1.4 shows that the set of vanishing loci of a spectra defines a topology on spectra.

Definition 1.1.5 (Zariski Topology on Spec) Let A be a commutative ring. Define the Zariski topology on Spec(A) to be the topology where the closed sets are exactly sets of the form $\mathbb{V}^{\mathrm{S}}(I)$ for $I \subseteq A$ an ideal of A.

Example 1.1.6 The Zariski topology on Spec$(\mathbb{Z})$ is given as follows.
- The points are given by Spec$(\mathbb{Z}) = \{(p) \mid p \text{ is a prime }\} \cup \{(0)\}$
- The closed subsets are of the form

$$\mathbb{V}^s((n)) = \{(p) \in \text{Spec}(\mathbb{Z}) \mid p \text{ divides } n\}$$

Example 1.1.7 The Zariski topology on Spec$(\mathbb{C}[x])$ is given as follows.
- The points are given by Spec$(\mathbb{C}[x]) = \{(x - a) \mid a \in \mathbb{C}\} \cup \{(0)\}$
- The closed subsets are precisely of the form

$$\mathbb{V}^s((p)) = \{(x - a) \mid a \text{ is a root of } p\}$$

for $p \in \mathbb{C}[x]$ since $\mathbb{C}[x]$ is a PID.

Definition 1.1.8 (Ideals from a Zero Locus)
Let R be a commutative ring. Let $S \subseteq$ Spec(R) be a subset. Define the ideal of associated to S to be

$$\mathbb{I}^{\mathrm{S}}(S) = \{f \in R \mid f \in P \text{ for all } P \in S\}$$

Lemma 1.1.9
Let R be a commutative ring. Then the following are true.
- If $S \subseteq R$ is a subset, then $\mathbb{I}^{\mathrm{S}}(S)$ is a radical ideal of R.
- If $S \subseteq T \subseteq R$, then $\mathbb{I}^{\mathrm{S}}(T) \subseteq \mathbb{I}^{\mathrm{S}}(S)$.
- If $S_1, S_2 \subseteq R$, then $\mathbb{I}^{\mathrm{S}}(S_1 \cup S_2) = \mathbb{I}^{\mathrm{S}}(S_1) \cap \mathbb{I}^{\mathrm{S}}(S_2)$.

Proof Clearly we have $\mathbb{I}^{\mathrm{S}}(S)=\bigcap_{P\in S}P$ and intersections of ideals is an ideal. Now let $f\in\sqrt{\mathbb{I}^{\mathrm{S}}(S)}$. Then $f^r\in\mathbb{I}^{\mathrm{S}}(S)$ for some $r\in\mathbb{N}$. This means that $f^r\in\bigcap_{P\in S}P$. Since all P are prime ideals, $f^r\in P$ implies $f\in P$. Hence $f\in\mathbb{I}^{\mathrm{S}}(S)$ and so $\mathbb{I}^{\mathrm{S}}(S)$ is radical.

Since $S\subseteq T$, we must have
$$\mathbb{I}^{\mathrm{S}}(T)=\bigcap_{P\in T}\subseteq\bigcap_{P\in S}P=\mathbb{I}^{\mathrm{S}}(S)$$
∎

Proposition 1.1.10 (Hilbert's Nullstellensatz for Spectra)
Let R be a commutative ring. Then the following are true.

- If I is an ideal of R, then $\mathbb{I}^{\mathrm{S}}(\mathbb{V}^{\mathrm{S}}(I))=\sqrt{I}$.
- If $S\subseteq\mathrm{Spec}(R)$ is a subset, then $\mathbb{V}^{\mathrm{S}}(\mathbb{I}^{\mathrm{S}}(S))=\overline{S}$.

Proof Let I be an ideal of R, then we have
$$\mathbb{I}^{\mathrm{S}}(\mathbb{V}^{\mathrm{S}}(I))=\bigcap_{P\in\mathbb{V}^{\mathrm{S}}(I)}P=\bigcap_{I\subseteq P\in\mathrm{Spec}(R)}P=\sqrt{I}$$

Now let $S\subseteq\mathrm{Spec}(R)$ be a subset. Notice that a closed subset $T=\mathbb{V}^{\mathrm{S}}(I)$ for some ideal $I\subseteq R$ contains S if and only if for all $P\in S$, $I\subseteq P$. And this is true if and only if $I\subseteq\mathbb{I}^{\mathrm{S}}(S)$. Since the largest ideal contained by $\mathbb{I}^{\mathrm{S}}(S)$ is itself, the smallest closed subset containing S must be $\mathbb{V}^{\mathrm{S}}(\mathbb{I}^{\mathrm{S}}(S))$. Hence $\overline{S}=\mathbb{V}^{\mathrm{S}}(\mathbb{I}^{\mathrm{S}}(S))$. ∎

Lemma 1.1.11
Let R be a commutative ring. Let $S\subseteq\mathrm{Spec}(R)$ be a subset. Then we have
$$\mathbb{I}^{\mathrm{S}}(\overline{S})=\mathbb{I}^{\mathrm{S}}(S)$$

Proof Since $S\subseteq T$, we must have
$$\mathbb{I}^{\mathrm{S}}(T)=\bigcap_{P\in T}\subseteq\bigcap_{P\in S}P=\mathbb{I}^{\mathrm{S}}(S)$$

We have
$$\mathbb{I}^{\mathrm{S}}(\overline{S})=\mathbb{I}^{\mathrm{S}}(\mathbb{V}^{\mathrm{S}}(\mathbb{I}^{\mathrm{S}}(S)))=\mathbb{I}^{\mathrm{S}}(S)$$
where the first equality comes from item 4 and the second equality comes from item 3. ∎

The lemma shows that we only need to concern ourselves with closed subsets of $\mathrm{Spec}(R)$ as the input of $\mathbb{I}^{\mathrm{S}}(-)$.

Proposition 1.1.12
Let R be a commutative ring. Then the following are true.

- There is an inclusion reversing bijection
$$\begin{Bmatrix}\text{Radical ideals}\\\text{of }R\end{Bmatrix}\overset{1:1}{\longleftrightarrow}\begin{Bmatrix}\text{Closed Subsets of}\\\mathrm{Spec}(R)\end{Bmatrix}$$
induced by $\mathbb{V}^{\mathrm{S}}(-)$ and $\mathbb{I}^{\mathrm{S}}(-)$.
- There is an inclusion reversing bijection
$$\mathrm{Spec}(R)\overset{1:1}{\longleftrightarrow}\begin{Bmatrix}\text{Irreducible Closed Subsets}\\\text{of }\mathrm{Spec}(R)\end{Bmatrix}$$
induced by $\mathbb{V}^{\mathrm{S}}(-)$ and $\mathbb{I}^{\mathrm{S}}(-)$.
- There is an inclusion reversing bijection
$$\begin{Bmatrix}\text{Minimal prime}\\\text{ideals of }R\end{Bmatrix}\overset{1:1}{\longleftrightarrow}\begin{Bmatrix}\text{Irreducible Components}\\\text{of }\mathrm{Spec}(R)\end{Bmatrix}$$

induced by $\mathbb{V}^{\mathrm{S}}(-)$ and $\mathbb{I}^{\mathrm{S}}(-)$.

Proof
The first inclusion reversing bijection comes from 1.1.9.

Let $P \in \operatorname{Spec}(R)$ be a point. I claim that $\mathbb{V}^{\mathrm{S}}(P)$ is irreducible. Indeed, $\mathbb{V}^{\mathrm{S}}(P) = \operatorname{Spec}(R/P)$ and R/P being an integral domain implies that $\mathbb{V}^{\mathrm{S}}(P)$ is irreducible. Conversely, let S be an irreducible closed subset of $\operatorname{Spec}(R)$. I claim that $\mathbb{I}^{\mathrm{S}}(S)$ is a point. Since S is closed, $S = \mathbb{V}^{\mathrm{S}}(I)$ for some radical ideal I of R, and so $\mathbb{I}^{\mathrm{S}}(S) = I$. I claim that I is a prime ideal. Let $fg \in I$. Then $S = \mathbb{V}^{\mathrm{S}}(I) \subseteq \mathbb{V}^{\mathrm{S}}(fg) = \mathbb{V}^{\mathrm{S}}(f) \cup \mathbb{V}^{\mathrm{S}}(g)$ and we have

$$S = \left(\mathbb{V}^{\mathrm{S}}(f) \cap S\right) \cup \left(\mathbb{V}^{\mathrm{S}}(g) \cap S\right)$$

Since S is irreducible, without loss of generality suppose that $S = \mathbb{V}^{\mathrm{S}}(f) \cap S$. Let $P \in S = \mathbb{V}^{\mathrm{S}}(I)$, then $P \in \mathbb{V}^{\mathrm{S}}(f)$ implies that $f \in P$. Thus we have $f \in \bigcap_{P \supseteq I} P = \sqrt{I} = I$. Hence I is prime.

Since $\mathbb{V}^{\mathrm{S}}$ and $\mathbb{I}^{\mathrm{S}}$ are inclusion reversing, the smallest primes ordered by inclusion is in bijection to the largest irreducible closed subsets ordered by inclusion. ■

This shows that each irreducible closed subset has exactly one generic point, and each irreducible closed subset can be written uniquely as the closure of some point in the spectrum.

1.2 Topological Properties of Spectra

Lemma 1.2.1 Let k be an algebraically closed field. Let V be an affine variety. Then the bijection given by the Hilbert's nullstellensatz

$$V \cong \operatorname{maxSpec}(k[V])$$

is a homeomorphism where $\operatorname{maxSpec}(k[V])$ inherits the subspace topology of $\operatorname{Spec}(k[V])$.

Let k be an algebraically closed field. For a closed subset $V \subseteq \mathbb{A}_k^n$, we have the following.

- There is a bijection
$$V \overset{1:1}{\longleftrightarrow} \operatorname{maxSpec}(V) \subseteq \operatorname{Spec}(V)$$
The geometry of V is enhanced because we have more points in general, given by the irreducible subvarieties.
- In the Zariski topology of V, closed subsets are affine subvarieties of V. In the Zariski topology of $\operatorname{Spec}(V)$, closed subsets are collections of irreducible subvarieties.
- For a closed subset V of $\operatorname{Spec}(k[V])$, we have $C = \mathbb{V}^s(I)$ for some $I \subseteq k[V]$ an ideal. There is a bijection of sets
$$C = \mathbb{V}^s(I) \overset{1:1}{\longleftrightarrow} \{W \subseteq \mathbb{V}(I) \mid W \text{ is irreducible }\}$$
- In particular, for an irreducible polynomial $f \in k[x_1, \dots, x_n]$, $(f) \in \operatorname{Spec}(k[V])$ is a point. The closure of (f) is given by
$$\overline{(f)} = \{m \in \operatorname{maxSpec}(k[V]) \mid (f) \subseteq m\} \cup \{(f)\}$$

Proposition 1.2.2
Let R be a commutative ring. Let $I \subseteq R$ be an ideal of R. Then there is natural homeomorphism

$$\operatorname{Spec}(R/I) = \mathbb{V}^{\mathrm{S}}(I)$$

induced by the correspondence theorem for ideals.

Proof
By the correspondence theorem for ideals, the two sets are equal. Now let $\mathbb{V}^{\mathsf{S}}(J/I)$ be a closed subset of $\text{Spec}(R/I)$ where J is an ideal of R (all ideals of R/I are of this form by the correspondence theorem). Then we have

$$\mathbb{V}^{\mathsf{S}}(T) = \{P \in \text{Spec}(R/I) \mid T \subseteq P\} \overset{1:1}{\longleftrightarrow} \{P \in \text{Spec}(R) \mid J \subseteq P\} = \mathbb{V}^{\mathsf{S}}(J)$$

Thus the closed sets of the two spaces coincide via the same bijection, namely that given by the correspondence theorem. ■

We can also explicitly write out the open sets and a basis for the Zariski topology.

Definition 1.2.3 (Distinguished Open Sets)
Let R be a commutative ring. Let $S \subseteq R$ be a subset. Define the distinguished open set of S to be

$$D(S) = \text{Spec}(R) \setminus \mathbb{V}^S(S)$$

If $S = \{f\}$, then we say that $D(S)$ is a basic open set.

Lemma 1.2.4
Let R be a commutative ring. Then the basic open sets form a basis for the Zariski topology on $\text{Spec}(R)$.

Proof
Let $S \subseteq R$ be a subset. Then we have

$$\begin{aligned} D(S) &= \text{Spec}(R) \setminus \mathbb{V}^{\mathsf{S}}(S) \\ &= \text{Spec}(R) \setminus \mathbb{V}^{\mathsf{S}}\left(\sum_{f \in S}(f)\right) \\ &= \text{Spec}(R) \setminus \bigcap_{f \in S} \mathbb{V}^{\mathsf{S}}((f)) \\ &= \bigcup_{f \in S}(D(f)) \end{aligned}$$

and so we are done. ■

Lemma 1.2.5
Let R be a commutative ring. Let $f \in R$. Then there is a natural homeomorphism

$$D(f) = \text{Spec}(R_f)$$

induced by the correspondence theorem for prime ideals of localization.

Proof By the correspondence of prime ideals of localization, we have

$$\text{Spec}(R_f) \overset{1:1}{\longleftrightarrow} \{P \in \text{Spec}(R) \mid P \cap \{f^n \mid n \in \mathbb{N}\}\} = D(f)$$

so we are done. Moreover, for any $T \subseteq R$ and $f \notin T$, we have

$$D(f) \supseteq \mathbb{V}^{\mathsf{S}}((T)) = \{P \in \text{Spec}(R_f) \mid (T)_f \subseteq P\} = \mathbb{V}^{\mathsf{S}}((T)_f)$$

■

We will prove some topological properties of spectra. Recall the following notions.

- A space X is irreducible if it is not the union of two proper closed subsets. Every space can be uniquely decomposed into its irreducible components, all of which are closed.

- A space X is connected if it is not the disjoint union of two open sets. Every space can be uniquely decomposed into its connected components, all of which are closed (but not necessarily open in X). Moreover, irreducible spaces are clearly connected (but the converse is not true).
- A space X is compact if any open cover has a finite subcover.
- A space X is Noetherian if for any descending chain of closed sets $Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_k \supseteq \cdots$, there exists $n \in \mathbb{N}$ such that $Z_n = Z_{n+1} = Z_{n+2} = \cdots$. The Noetherianess property guarantees that the decomposition into irreducible components is finite.

Proposition 1.2.6

Let R be a commutative ring. Then the following are true.

- R is an integral domain if and only if Spec(R) is irreducible.
- Spec(R) is connected if and only if R is not a product of rings.
- Spec(R) is compact.
- If R is Noetherian, then Spec(R) is Noetherian.

Proof

- Suppose that Spec(R) is reducible. Then there exists ideals $I, J \neq (0)$ of R such that Spec$(R) = \mathbb{V}^{\mathsf{S}}(I) \cup \mathbb{V}^{\mathsf{S}}(J) = \mathbb{V}^{\mathsf{S}}(I \cap J)$. Since $P \supseteq I \cap J$ for all prime ideals P of R, we have $(0) \neq IJ \subseteq I \cap J \subseteq N(R)$. Thus $N(R)$ is non-zero and so R is not an integral domain.

  Now suppose that Spec(R) is irreducible. Let $f, g \in R$ such that $fg = 0$. Then we have
  $$\mathbb{V}^{\mathsf{S}}((f)) \cup \mathbb{V}^{\mathsf{S}}((g)) = \mathbb{V}^{\mathsf{S}}((fg)) = \mathbb{V}^{\mathsf{S}}((0)) = \text{Spec}(R)$$
  Since Spec(R) is irreducible, without loss of generality suppose that $\mathbb{V}^{\mathsf{S}}((f)) = \text{Spec}(R)$. Then $f \in P$ for all prime ideals P of R. Hence $f \in N(R) = \sqrt{(0)}$, so $f^n = 0$ and $f = 0$.
- Suppose that $R = S_1 \times S_2$ is a product of rings. Then there is a natural isomorphism Spec$(R) \cong \text{Spec}(S_1) \amalg \text{Spec}(S_2)$ induced by the correspondence theorem for ideals. Hence R is not connected. Suppose that R is not connected. Then there exists non trivial proper open subsets $D(I), D(J)$ such that Spec$(R) = D(I) \amalg D(J) = \mathbb{V}^{\mathsf{S}}(J) \amalg \mathbb{V}^{\mathsf{S}}(I)$.
- Since $\{D(f) \mid f \in R\}$ is a basis for the Zariski topology, it suffices to consider an open cover consisting of these basis elements. So let $\{D(f_i) \mid i \in I\}$ be an open cover of Spec(R). Then we have Spec$(R) = \bigcup_{i \in I} D(f_i)$ implies
  $$\emptyset = \bigcap_{i \in I} \mathbb{V}^{\mathsf{S}}(f_i)$$
  This means that for all $P \in \text{Spec}(R)$, there exists $i \in I$ such that $f_i \notin P$ and $(f_i) \not\subseteq P$. So there is no prime ideal P that contains $\sum_{i \in I}(f_i)$. In particular, there is no maximal ideal that contains this ideal. Hence $\sum_{i \in I}(f_i) = (1)$. By definition of sums of ideals, there exists $x_i \in R$ non-zero for all but finitely many, such that $\sum_{i \in I} x_i f_i = 1$. So there exists $x_1, \dots, x_n \in R$ and $f_1, \dots, f_n$ such that $\sum_{i=1}^n x_i f_i = 1$.

  Suppose for a contradiction that $\emptyset \neq \bigcap_{i=1}^n \mathbb{V}^{\mathsf{S}}(f_i)$, then there exists a prime ideal P of R such that $f_i \in P$. Hence $\sum_{i=1}^n (f_i) \subseteq P$. But since $\sum_{i=1}^n x_i f_i = 1$, we have $1 \in P$ and this is a contradiction. Hence we must have $\emptyset = \bigcap_{i=1}^n \mathbb{V}^{\mathsf{S}}(f_i)$. In other words, Spec$(R) = \bigcup_{i=1}^n D(f_i)$ and so we have found a finite subcover for the given open cover.
- Suppose that R is Noetherian. Let
  $$\mathbb{V}^{\mathsf{S}}(I_1) \supseteq \mathbb{V}^{\mathsf{S}}(I_2) \supseteq \cdots \supseteq \mathbb{V}^{\mathsf{S}}(I_k) \supseteq \cdots$$
  be a decreasing sequence of closed subsets of Spec(R). Then applying the inclusion reversing bijection $\mathbb{I}^{\mathsf{S}}(-)$ gives an ascending chain of ideals
  $$\sqrt{I_1} \subseteq \sqrt{I_2} \subseteq \cdots \subseteq \sqrt{I_k} \subseteq \cdots$$

of R. Since R is Noetherian, there exists $n \in \mathbb{N}$ such that $\sqrt{I_n} = \sqrt{I_{n+t}}$ for all $t \in \mathbb{N}$. This means that
$$\mathbb{V}^{\mathsf{S}}(I_n) = \mathbb{V}^{\mathsf{S}}(\sqrt{I_n}) = \mathbb{V}^{\mathsf{S}}(\sqrt{I_{n+t}}) = \mathbb{V}^{\mathsf{S}}(I_{n+t})$$
Thus Spec(R) is Noetherian.

■

1.3 The Structure Sheaf of a Spectrum

We now define the structure sheaf on a spectrum so that they form a locally ringed space.

Definition 1.3.1 (Structure Sheaf)
Let R be a commutative ring and Spec(R) the spectrum of A as a topological space. Define the structure sheaf
$$\mathcal{O}_{\text{Spec}(R)} : \textbf{Open}(\text{Spec}(R)) \to \textbf{Rings}$$
on Spec(R) by the following.

- For each $U \subseteq X$ open, define
$$\mathcal{O}_{\text{Spec}(R)}(U) = \left\{ (s_p)_{p\in U} \in \prod_{p\in U} R_p \;\middle|\; \substack{\forall p, \exists p \in U_p \subseteq U \text{ open and } a, f \in R \text{ s.t.} \\ q \in U_p \text{ implies } f \notin q \text{ and } s_q = a/f \in A_q} \right\}$$
- For $V \subseteq U$ an inclusion, the unique morphism $\mathcal{O}_{\text{Spec}(R)}(U) \to \mathcal{O}_{\text{Spec}(R)}(V)$ is the projection map
$$\text{res}^U_V : \prod_{p\in U} R_p \to \prod_{p\in V} R_p$$

Proposition 1.3.2
Let R be a commutative ring. Then the structure sheaf
$$\mathcal{O}_{\text{Spec}(R)} : \textbf{Open}(\text{Spec}(R)) \to \textbf{Set}$$
defined above is indeed a sheaf on Spec(R).

Proof
Identity: Let $U \subseteq \text{Spec}(R)$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U. Suppose that $(s_p)_{p\in U}, (t_p)_{p\in U} \in \mathcal{O}_{\text{Spec}(R)}(U)$ are such that $(s_p)_{p\in U_i} = (t_p)_{p\in U_i}$ for all $i \in I$. Since the U_i cover U, we check that for all $p \in U$, we have $s_p = t_p$ so that $(s_p)_{p\in U} = (t_p)_{p\in U}$.

Gluing: Let $U \subseteq \text{Spec}(R)$ be an open set. Let $\{U_i \mid i \in I\}$ be an open cover of U. Suppose that $((s_i)_p)_{p\in U_i}$ for ranging i is a collection of sections such that $((s_i)_p)_{p\in U_i\cap U_j} = ((s_j)_p)_{p\in U_i\cap U_j}$. Define a section in $\mathcal{O}_{\text{Spec}(R)}(U)$ by the collection $t_p = (s_i)_p$ if $p \in U_i$. Since the collection $((s_i)_p)_{p\in U_i}$ agrees on intersections, $(t_p)_{p\in U}$ is well defined, and moreover restricts to $((s_i)_p)_{p\in U_i}$ for each i. Hence we are done. ■

The structure sheaf allows Spec(R) to be a ringed space. The ringed space on any spectrum is in fact a locally ringed space. But this is not true for the ringed space on varieties in the classical sense.

Proposition 1.3.3
Let R be a commutative ring. Then the following are true regarding the ringed space $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$.

- For any $p \in \text{Spec}(R)$, there is a natural isomorphism $\mathcal{O}_{\text{Spec}(R),p} \cong R_p$ on the level of stalks.
- $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ is a locally ringed space.
- For any element $f \in R$, there is an isomorphism $\mathcal{O}_{\text{Spec}(R)}(D(f)) \cong R_f$
- There is an isomorphism $\mathcal{O}_{\text{Spec}(R)}(\text{Spec}(R)) \cong R$ on the global level.

Proof

- Let $p \in \text{Spec}(R)$. For each open set $U \subseteq X$ containing p, there is an evident ring homomorphism $\mathcal{O}_{\text{Spec}}(R)(U) \to R_p$ that is projection. By the universal property of colimits, this induces a ring homomorphism $\mathcal{O}_{\text{Spec}(R),p} \to R_p$ defined by $[((s_q)_{q\in V}, V)] \mapsto s_p$. It remains to show that this map is a bijection. Suppose that $[((s_q)_{q\in V}, V)]$ is such that $s_p = 0 \in R_p$. Then there exists an open subset $p \in W \subseteq V$ and $f, g \in R$ such that for all $q \in W$, $g \notin q$ and $s_q = f/g$. In particular this means that $s_p = f/g = 0$. This is equivalent to there exists $h \in R \setminus p$ such that $hf = 0$. Let $S = D(g) \cap D(h) \cap W$. Notice that $p \in S$. Let $q \in S$. In R_q, we have $g, h \notin q$ and so $hf = 0$ implies that $0 = f/g = s_q$ in R_q. Hence $(s_q)_{q\in S} = 0$. Then we have $[((s_q)_{q\in V}, V)] = [((s_q)_{q\in S}, S)] = 0$ so that the ring homomorphism is injective. For surjectivity, let $s_p \in R_p$. Then there exists $f, g \in R$ and $g \notin p$ such that $s_p = f/g$. Define a section $(f/g)_{p\in D(g)} \in \mathcal{O}_{\text{Spec}(R)}(D(g))$. It is a well defined element because on the open set $D(g)$, for any $q \in D(g)$ we clearly have $s_q = f/g \in R_q$ so that the compatibility requirement is satisfied. Evidently, we have $[((f/g)_{p\in D(g)}, D(g))] \in \mathcal{O}_{\text{Spec}(R),p}$ is mapped to $s_p = f/g \in R_p$. Hence the ring homomorphism is surjective.
- From the above we immediately conclude that every stalk of the ringed space is a local ring. Hence $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ is a locally ringed space.
- Define a map $\phi : R_f \to \mathcal{O}_{\text{Spec}(R)}(D(f))$ by $\phi(r/f^n) = (r/f^n)_{q\in D(f)}$. It is clear that it is a ring homomorphism. It remains to show that the map is bijective. Suppose that $(r/f^n)_{q\in D(f)} = (s/f^m)_{q\in D(f)}$. Then for each $q \in D(f)$, we have $r/f^n = s/f^m \in R_q$. This means that there exists $h \in R \setminus q$ such that $h(rf^m - sf^n) = 0$. Now notice that $h \in \text{Ann}_R(rf^m - sf^n)$ and $h \notin q$. Hence $\text{Ann}_R(rf^m - sf^n) \notin q$. Since this is true for all $q \in D(f)$, we must have

$$\mathbb{V}^{\text{S}}(\text{Ann}_R(rf^m - sf^n)) \cap D(f) = \emptyset$$
$$\mathbb{V}^{\text{S}}(\text{Ann}_R(rf^m - sf^n)) \setminus \mathbb{V}^{\text{S}}((f)) = \emptyset$$

and so $\mathbb{V}^{\text{S}}(\text{Ann}_R(rf^m - sf^n)) \subseteq \mathbb{V}^{\text{S}}((f))$. Taking $\mathbb{I}^{\text{S}}(-)$ gives $(f) \subseteq \sqrt{(f)} \subseteq \sqrt{\text{Ann}_R(rf^m - sf^n)}$. Then $f \in (f)$ means $f \in \sqrt{\text{Ann}_R(rf^m - sf^n)}$ so there exists $k \in \mathbb{N}$ for which $f^k(rf^m - sf^n) = 0$. Thus we have $r/f^n = s/f^m$ in R_f. Hence our map is injective.

For surjectivity, let $(s_p)_{p\in D(f)} \in \mathcal{O}_{\text{Spec}(R)}(D(f))$. For each $p \in D(f)$, there exists an open neighbourhood $V_p \subseteq D(f)$ of p and $r_p, g_p \in R$, $g_p \notin p$ such that $s_q = r_p/g_p$ for all $q \in V_p$. Notice that the open sets $\{D(g_p) \mid p \in D(f)\}$ cover $D(f)$. Now I claim that this open cover has a finite subcover. Indeed, we know that $D(f) \subseteq \bigcup_{p\in D(f)} D(g_p)$. Taking complements, we have $\bigcap_{p\in D(f)} \mathbb{V}^{\text{S}}(g_p) \subseteq \mathbb{V}^{\text{S}}(f)$. Notice that the intersection is equal to $\mathbb{V}^{\text{S}}(\sum_{p\in D(f)}(g_p))$. Taking $\mathbb{I}^{\text{S}}(-)$ on both sides give $\sqrt{(f)} \subseteq \sqrt{\sum_{p\in D(f)}(g_p)}$. So there exists $k \in \mathbb{N}$ such that $f^k \in \sum_{p\in D(f)}(g_p)$. By definition of sum of ideals, there exists a finite indexing set I and $a_i \in R$ such that $f^k = \sum_{i\in I} a_i g_i$. Then $f^k \in \sum_{i\in I}(g_i)$, and taking $\mathbb{V}^{\text{S}}(-)$ gives $\bigcap_{i\in I} \mathbb{V}^{\text{S}}(g_i) \subseteq \mathbb{V}^{\text{S}}(f)$ and taking complements give $D(f) \subseteq \bigcup_{i\in I} D(g_i)$. Hence $D(f) \subseteq D(g_1) \cup \cdots \cup D(g_r)$ has a finite subcover. On the open set $D(g_i g_j) = D(g_i) \cap D(g_j)$, for any $q \in D(g_i g_j)$, $s_q = r_i/g_i$ on $D(g_i)$ and $s_q = r_j/g_j$ on $D(g_j)$ so $r_i/g_i = s_q = r_j/g_j$ in $D(g_i g_j)$. Since we have just proven that the map $R_{g_i g_j} \to \mathcal{O}_{\text{Spec}(R)}(D(g_i g_j))$ is injective, we have $r_i/g_i = r_j/g_j \in R_{g_i g_j}$. So there exists $n_{i,j} \in \mathbb{N}$ such that $(g_i g_j)^{n_{i,j}}(r_i g_j - r_j g_i) = 0$. Let $n = \max\{n_{i,j}, 1 \leq i, j \leq r\}$. Then for all $1 \leq i, j \leq r$, we have $(g_i g_j)^n(r_i g_j - r_j g_i) = 0$. Let $r_i' = r_i g_i^n$ and $g_i' = g_i^{n+1}$. Then rewriting the equation $(g_i g_j)^n(r_i g_j - r_j g_i) = 0$ gives

$$g_j' r_i' - g_i' r_j' = 0$$

Notice that we still have $s_q = \frac{r_i'}{g_i'}$ for all $q \in D(g_i)$. Now since $D(g_i) = D(g_i')$ for $i \in I$ covers $D(f)$, there exists $k \in \mathbb{N}$ and $a_i \in R$ such that $f^k = \sum_{i\in I} a_i g_i'$ similar to what we did above to find the finite subcover. Let $r = \sum_{i\in I} a_i r_i'$. Then we have

$$g_j' r = \sum_{i\in I} a_i r_i' g_j' = \sum_{i\in I} a_i r_j' g_i' = f^k r_j'$$

Then for all $j \in I$, we have $\frac{r}{f^k} = \frac{r_j}{g_j'}$ on $D(g_j) = D(g_j')$. Thus we have that the element $\frac{r}{f^k}$ maps to our given element. Thus we are done.

- Apply the above by $f = 1_R$.

1.4 The Spec Functor

Definition 1.4.1 (Induced Map on Spectrum) Let R, S be commutative rings. Let $\phi : R \to S$ be a ring homomorphism. Define the function

$$\phi^\flat : \text{Spec}(S) \to \text{Spec}(R)$$

by the formula $P \mapsto \phi^{-1}(P)$

Proposition 1.4.2 Let A, B, C be commutative rings. Let $\phi : A \to B$ and $\psi : B \to C$ be ring homomorphisms. Then the following are true.

- $(\psi \circ \phi)^\flat = \phi^\flat \circ \psi^\flat$
- $(\text{id}_A)^\flat = \text{id}_{\text{Spec}(A)}$

Lemma 1.4.3
Let R, S be commutative rings. Let $\phi : R \to S$ be a ring homomorphism. Then the induced map

$$\phi^\flat : \text{Spec}(S) \to \text{Spec}(R)$$

is continuous.

Proof
Let $T = \mathbb{V}^S(I)$ be a closed set in $\text{Spec}(R)$. Then $(\phi^\flat)^{-1}(T) = \{P \in \text{Spec}(S) \mid \phi^{-1}(P) = \phi^\flat(P) \in T\}$. I claim that this set is equal to $\mathbb{V}^S(\phi(I))$. Let $P \in (\phi^\flat)^{-1}(T)$. Then $\phi^{-1}(P) \in T$ implies $I \subseteq \phi^{-1}(P)$ which means that $\phi(I) \subseteq P$. Hence $P \in \mathbb{V}^S(\phi(I))$. Now suppose that $P \in \mathbb{V}^S(\phi(I))$. Then $\phi(I) \subseteq P$ implies that $I \subseteq \phi^{-1}(\phi(I)) \subseteq \phi^{-1}(P)$. Hence $\phi^{-1}(P) \in T$. Thus the preimage of a closed set is closed.

Proposition 1.4.4 Let V, W be affine varieties over $\mathbb{C}$. Let $F : V \to W$ be a morphism of affine varieties. Let m_x be the maximal ideal of $\mathbb{C}[V]$ corresponding to the point x. Then

$$(F^*)^\flat(m_x) = m_{F(x)}$$

We thus now have the following nice diagram:

$$\begin{tikzcd}
\text{maxSpec}(\mathbb{C}[V]) \arrow[r, "\phi^\flat|_{\text{maxSpec}(\mathbb{C}[V])}"] \arrow[d, leftrightarrow, "1:1"'] & \text{maxSpec}(\mathbb{C}[W]) \arrow[d, leftrightarrow, "1:1"] \\
V \arrow[r, "\phi"'] & W
\end{tikzcd}$$

Proposition 1.4.5 Let V be an affine variety over $\mathbb{C}$. Let $X \subseteq V$ be an affine irreducible subvariety. Let P_X be the prime ideal of $\mathbb{C}[V]$ corresponding to X. Let $Y = \overline{F(X)}$. Then

$$(F^*)^\flat(P_X) = P_Y$$

Example 1.4.6 Let $\sigma : \mathbb{C}[x] \to \mathbb{C}[[t]]$ be a map. Let $\iota : \mathbb{C}[[t]] \to \mathbb{C}((t))$ be the inclusion map. Consider the following diagram:

$$\begin{tikzcd}
\mathbb{C}[x] \arrow[rr, "\sigma"] & & \mathbb{C}((t)) \\
& \mathbb{C}[[t]] \arrow[ur, hook, "\iota"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
\substack{\mathrm{Spec}(\mathbb{C}[x])= \\ \{(0)\}\cup\{(x-a)\mid a\in\mathbb{C}\}} & & \mathrm{Spec}(\mathbb{C}((t)))=\{(0)\} \arrow[ll, "\sigma^\flat"'] \arrow[dl, "\iota^\flat"] \\
& \mathrm{Spec}(\mathbb{C}[[t]])=\{(0),(t)\} \arrow[ul, dashed, "\rho"] &
\end{tikzcd}$$

- Let σ be the map $x \mapsto t$. Then ρ exists.
- Let σ be the map $x \mapsto t+1$. Then ρ exists.
- Let σ be the map $x \mapsto 1/t$. Then ρ does not exist.
- Let σ be the map $x \mapsto t^2$. Then ρ exists.
- Let σ be the map $x \mapsto e^t$. Then ρ exists.

Proof We know that the Spec functor is fully faithful. This means that ρ exists if and only if there exists a map $\mathbb{C}[x] \to \mathbb{C}[[t]]$ making the top diagram commute.

For the first two cases, the images of the map lie inside $\mathbb{C}[[t]]$. Hence the map $\mathbb{C}[x] \to \mathbb{C}[[t]]$ exists, and hence ρ exists. In the first case, we have $\rho((t)) = (x)$ and in the second case we have $\rho((t)) = (x-1)$.

For the third case, no map makes the top diagram commute hence ρ does not exist.

For the fourth and fifth cases, the images of the map lie inside $\mathbb{C}[[t]]$. Hence ρ exists. In the fourth case we have $\rho((t)) = (x)$ and in the fifth case we have $\rho((t)) = (x-1)$. Indeed, $f \in \rho^{-1}(t)$ if and only if t divides $f(e^t)$. This means that $g(t) = f(e^t)$ has no constant term, or that $g(0) = 0$. This means that $f(1) = 0$, hence $x-1$ divides f. ■

Definition 1.4.7 (The Spec Functor) Define the spec functor

$$\mathrm{Spec} : \mathbf{CRing} \to \mathbf{LRS}$$

to consist of the following data.

- A commutative ring A is sent to the locally ringed space $(\mathrm{Spec}(A), \mathcal{O}_{\mathrm{Spec}(A)})$.
- For a ring homomorphism $\phi : A \to B$, we have that $\mathrm{Spec}(\phi)$ consists of a continuous map $\phi^\flat : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ and a morphism of sheaves

$$(\phi^\flat)^\# : \mathcal{O}_{\mathrm{Spec}(A)} \to (\phi^\flat)_*\mathcal{O}_{\mathrm{Spec}(B)}$$

defined as follows. Fixing $p \in \mathrm{Spec}(A)$, for each $q \in (\phi^\flat)^{-1}(p)$, ϕ induces a local homomorphism $\phi_{p,q} : A_p \to B_q$. Ranging over q, we obtain a map

$$\phi_p : A_p \to \prod_{q\in(\phi^\flat)^{-1}(p)} B_q$$

For $U \subseteq \mathrm{Spec}(A)$ an open set, ranging over $p \in U$ gives a map

$$\prod_{p\in U} \phi_p : \prod_{p\in U} A_p \to \prod_{q\in(\phi^\flat)^{-1}(U)} B_q$$

The morpnism then sends $f \in \mathcal{O}_{\mathrm{Spec}(A)}(U)$ to the composite

$$(\phi^\flat)^{-1}(U) \xrightarrow{\phi^\flat} U \xrightarrow{f} \prod_{p\in U} A_p \xrightarrow{\prod_{p\in U}\phi_p} \prod_{q\in(\phi^\flat)^{-1}(U)} B_q$$

Lemma 1.4.8
Let R, S be commutative rings. Let $\varphi : R \to S$ be a ring homomorphism. Let $f \in R$. Then the morphism

$$R_f \cong \mathcal{O}_{\text{Spec}(R)}(D(f)) \to \mathcal{O}_{\text{Spec}(S)}((\varphi^\flat)^{-1}(D(f))) = \mathcal{O}_{\text{Spec}(S)}(D(\varphi(f))) \cong S_{\varphi(f)}$$

is the induced morphism of localization $\varphi_f : R_f \to S_{\varphi(f)}$.

Proof
For any $p \in D(f)$ and $q \in (\varphi^\flat)^{-1}(p)$, notice that the localization morphism $\varphi_p : R_p \to S_q$ is defined by $r/s \mapsto \varphi(r)/\varphi(s)$. Since the following diagram commtues

$$\begin{tikzcd} R_f \arrow[r, "\frac{r}{f^n} \mapsto \frac{\varphi(r)}{\varphi(f)^n}"] \arrow[d, "\cong"] & S_{\varphi(f)} \arrow[d, "\cong"'] \\ \mathcal{O}_{\text{Spec}(R)}(D(f)) \arrow[r, "\left(\frac{r}{f^n}\right)_{p \in D(f)} \mapsto \left(\frac{\varphi(r)}{\varphi(f)^n}\right)_{q \in D(\varphi(f))}"] & \mathcal{O}_{\text{Spec}(S)}(D(\varphi(f))) \end{tikzcd}$$

by naturality of the vertical isomorphisms, we check that our morphism is indeed in its desired form. ■

Proposition 1.4.9
The spec functor is fully faithful.

Proof
Clearly the functor is injective by since there is a natural isomorphism $\mathcal{O}_{\text{Spec}(R)}(\text{Spec}(R)) \cong R$. For surjectivity, suppose that $(\varphi, \varphi^\#) : (\text{Spec}(S), \mathcal{O}_{\text{Spec}(S)}) \to (\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ is a morphism of spectra. Let $q \in \text{Spec}(S)$. By the universal property of colimits and 1.3.3, the following diagram on the left can be simplified to the diagram on the right:

$$\begin{tikzcd} \mathcal{O}_{\text{Spec}(R)}(\text{Spec}(R)) \arrow[r, "\varphi^\#(\text{Spec}(R))"] \arrow[d] & \mathcal{O}_{\text{Spec}(S)}(S) \arrow[d] \\ \mathcal{O}_{\text{Spec}(R),\varphi(q)} \arrow[r] & \mathcal{O}_{\text{Spec}(S),q} \end{tikzcd} \qquad \begin{tikzcd} R \arrow[r] \arrow[d] & S \arrow[d] \\ R_{\varphi(q)} \arrow[r] & S_q \end{tikzcd}$$

By the universal property of localization, the morphism $R_{\varphi(q)} \to S_q$ allowing the right diagram to commute is unique. Hence the ring homomorphism between stalks is precisely the morphism induced by the universal property of localization. Now since q is the unique maximal ideal of S_q and $\varphi(q)$ is the unique maximal ideal of $R_{\varphi(q)}$, we know that this homomorphism must send the maximal ideal $\varphi(q)$ of $R_{\varphi(q)}$ to q of S_q. By the natural bijection between prime ideals of a ring and its localization, $\varphi(q)$ corresponds to the same prime ideal in R and q corresponds to the same prime ideal in S. Moreover, the since the diagram is commutative, the map $\varphi^\#(\text{Spec}(R)) : R \to S$ must send $\varphi(q)$ to q.

I claim that the preimage of $(\varphi, \varphi^\#)$ is precisely the morphism $\varphi^\#(\text{Spec}(R)) : R \to S$. Indeed, the induced morphism $\text{Spec}(\varphi^\#)$ sends q to $(\varphi^\#)^{-1}(q)$. But we have just established above that $\varphi^\#$ sends $\varphi(q)$ to q. Hence $\text{Spec}(\varphi^\#)$ must send q to $\varphi(q)$. Thus $\text{Spec}(\varphi^\#)$ agrees with $(\varphi, \varphi^\#)$ on the level of topological spaces.

It remains to show that the two map of sheaves are equal. To do this, it suffices to show that the induced map of stalks are equal. Notice that the two map of sheaves are equal in their global section and their map of topological spaces, hence each of the two map of sheaves induces morphisms $f, g : R_{\varphi(q)} \to S_q$ making the above diagram commute. But we have established that such a morphism allowing the above diagram to commute must be induced by the universal property of localization, and so must be unique. Hence the induced morphism of stalks of the two map of sheaves are equal. Thus we are done. ■

1.5 The Associated Sheaf of Modules

We restate the definition here of a sheaf of modules here. Let $\mathcal{A}$ be a sheaf of rings over X. Let U be an open set of X. A sheaf of $\mathcal{A}$-modules over X is a sheaf $\mathcal{F}$ such that each $\mathcal{F}(U)$ is an $\mathcal{A}(U)$-modules. Moreover, for each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathcal{F}(U) \to \mathcal{F}(V)$ is such that the following diagram commutes:

$$\begin{tikzcd} \mathcal{A}(U) \times \mathcal{F}(U) \arrow[r, "\text{action}"] \arrow[d, "\text{res}_{U,V} \times \text{res}_{U,V}"'] & \mathcal{F}(U) \arrow[d, "\text{res}_{U,V}"] \\ \mathcal{A}(V) \times \mathcal{F}(V) \arrow[r, "\text{action}"] & \mathcal{F}(V) \end{tikzcd}$$

For a scheme $(X, \mathcal{O}_X)$ that is also a locally ringed space, we denote the category of sheaves of $\mathcal{O}_X$-modules as $\text{Mod}_{\mathcal{O}_X}$.

It is the analogue of a module over a ring for the following reason.

Definition 1.5.1 (Associated Sheaf)
Let R be a commutative ring. Let M be an R-module. Define a sheaf $\widetilde{M}$ on $\text{Spec}(R)$ as follows.

- For each open set $U \subseteq \text{Spec}(R)$, define

$$\widetilde{M}(U) = \left\{ (s_p)_{p \in U} \in \prod_{p \in U} M_p \;\middle|\; \begin{array}{c} \forall p, \exists p \in U_p \subseteq U \text{ open and } m \in M, f \in R \text{ s.t.} \\ q \in U_p \text{ implies } f \notin q \text{ and } s_q = m/f \in M_q \end{array} \right\}$$

- For $V \subseteq U$ an inclusion, define the unique morphism $\widetilde{M}(U) \to \widetilde{M}(V)$ given by restriction.

Lemma 1.5.2
Let M be an R-module. Then the associated sheaf is a sheaf of $\mathcal{O}_{\text{Spec}(R)}$-modules.

Proof
Same proof as 1.3.2. ■

Proposition 1.5.3
Let R be a commutative ring. Let M be an R-module. Then the following are true regarding the associated sheaf $\widetilde{M}$.

- For each $p \in \text{Spec}(R)$, there is a natural isomorphism $\widetilde{M}_p \cong M_p$
- For any $f \in R$, there is a natural isomorphism $\widetilde{M}(D(f)) \cong M_f$ of R_f-modules
- There is a natural isomorphism $\widetilde{M}(\text{Spec}(R)) \cong M$.

Proof Same proof as 1.3.3. ■

Definition 1.5.4 (The Functor of Associated Sheaf)
Let R be a commutative ring. Define the associated sheaf functor

$$\widetilde{(-)} : \textbf{Mod}_R \to \textbf{Mod}_{\mathcal{O}_{\text{Spec}(R)}}$$

as follows.

- For each R-module M, define $\widetilde{M}$ to be the associated sheaf.
- For each R-module homomorphism $\varphi : M \to N$, define the induced morphism $\widetilde{\varphi}$ on each open set $U \subseteq X$ as follows. For each $p \in U$, there is an induced R-module homomorphism $M_p \to N_p$. Define $\widetilde{\varphi}$ by the restriction of the product

$$\prod_{p \in U} M_p \to \prod_{p \in U} N_p$$

to $\widetilde{M}(U)$.

We have to check that the image of the restriction of the product lies inside $\widetilde{N}(U)$.

Proposition 1.5.5
Let R be a commutative ring. Then $\widetilde{(-)}$ is fully faithful.

Proof Let $\varphi, \psi : M \to N$ be R-module homomorphisms. Since the R-module isomorphism $M = M_1 \cong \widetilde{M}(\text{Spec}(R))$ is natural, we obtain a commutative diagram:

$$\begin{tikzcd}
M \arrow[r, "\varphi{,}\psi"] \arrow[d, "\cong"'] & N \arrow[d, "\cong"] \\
\widetilde{M}(\text{Spec}(R)) \arrow[r, "\widetilde{\varphi}=\widetilde{\psi}"'] & \widetilde{N}(\text{Spec}(R))
\end{tikzcd}$$

Hence $\varphi = \psi$. Thus $\widetilde{(-)}$ is faithful. Now let $f : \widetilde{M} \to \widetilde{N}$ be a morphism of $\mathcal{O}_{\text{Spec}}(R)$-modules. Consider the following diagram:

$$\begin{tikzcd}
 & & & M_p \arrow[dd, "f(\text{Spec}(R))_p"] \\
\widetilde{M}(\text{Spec}(R)) \arrow[urrr] \arrow[rr] \arrow[dr] \arrow[dd, "f(\text{Spec}(R))"'] & & \widetilde{M}_p \arrow[ur, dashed, "\exists!"] \arrow[dd, dashed, "\exists! f_p"] & \\
 & \widetilde{M}(U) \arrow[ur] \arrow[dd, "f(U)"] & & N_p \\
\widetilde{N}(\text{Spec}(R)) \arrow[urrr] \arrow[rr] \arrow[dr] & & \widetilde{N}_p \arrow[ur, dashed, "\exists!"'] & \\
 & \widetilde{N}(U) \arrow[ur] & &
\end{tikzcd}$$

It is commutative by the naturality of colimits and localization. Thus the morphisms $\widetilde{M}(U) \to \widetilde{N}(U)$ is given by the product of the localizations $f(\text{Spec}(R))_p$ for each $p \in U$. By construction of $\widetilde{(-)}$, the morphisms $f(\widetilde{\text{Spec}(R)})(U)$ is also given by the product of the localizations. Thus the morphisms $f(\widetilde{\text{Spec}(R})$ and f are the same. Thus the preimage of f is given by $f(\text{Spec}(R))$. Hence the functor $\widetilde{(-)}$ is full. ■

Proposition 1.5.6 Let R be a commutative ring. Then there is an adjunction

$$\widetilde{(\,\cdot\,)} : \mathbf{Mod}_R \rightleftarrows \mathbf{Mod}_{\mathcal{O}_{\text{Spec}(R)}} : \Gamma$$

Proposition 1.5.7
Let R be a commutative ring. Then $\widetilde{(-)}$ and Γ are exact functors.

2 Projective Spectra and its Structure Sheaf

2.1 The Zariski Topology and Hilbert's Nullstellensatz for Proj

Let $S = \bigoplus_{i=0}^\infty S_i$ be a commutative graded ring. Recall that the irrelevant ideal of S is defined by

$$S_+ = \bigoplus_{i=1}^\infty S_i$$

More generally, recall that we defined

$$I_+ = I \cap S_+$$

for a homogeneous ideal $I \subseteq S$ in Rings and Modules. It is also a homogeneous ideal.

Definition 2.1.1 (The Projective Spectrum) Let S be a commutative graded ring. Define the projective spectrum of S to be

$$\text{Proj}(S) = \{P \in \text{Spec}(S) \mid P \text{ is homogeneous and } S_+ \not\subseteq P\}$$

where S_+ is the irrelevant ideal.

Definition 2.1.2 (The Homogeneous Vanishing Set)
Let S be a commutative graded ring. Let I be a homogeneous ideal of S. Define the vanishing set of I to be

$$\mathbb{V}^H(I) = \{P \in \text{Proj}(S) \mid I \subseteq S\}$$

Lemma 2.1.3
Let S be a commutative graded ring. Then the following are true.

- If $I \subseteq J$ are homogeneous ideals of S, then $\mathbb{V}^H(J) \subseteq \mathbb{V}^H(I)$.

Lemma 2.1.4
Let S be a commutative graded ring. Let $I \subseteq S$ be a homogeneous ideal. Then $\mathbb{V}^H(I) = \emptyset$ if and only if I is an irrelevant ideal.

This justifies the terminology for irrelevant ideals: They have no geometric meaning at all in terms of the Proj construction.

Proposition 2.1.5
Let S be a commutative graded ring. Then the following are true.

- Let $\{I_j \mid j \in J\}$ be a set of homogeneous ideals of S. Then we have

$$\mathbb{V}^H\left(\sum_{j\in J} I_j\right) = \bigcap_{j\in J} \mathbb{V}^H(I_j)$$

- Let I_1, I_2 be homogeneous ideals of S. Then we have

$$\mathbb{V}^H(I_1 I_2) = \mathbb{V}^H(I_1) \cup \mathbb{V}^H(I_2)$$

Similar to that of $\text{Spec}(R)$ we can endow a topology on $\text{Proj}(S)$.

Definition 2.1.6 (Zariski Topology on Proj)
Let S be a commutative graded ring. Define the Zariski topology on $\text{Proj}(S)$ to be the topology where the closed sets are exactly sets of the form $\mathbb{V}^H(I)$ for $I \subseteq S$ a homogeneous ideal.

Lemma 2.1.7

Let S be a commutative graded ring. Let I be a homogeneous ideal of S. Then we have

$$\mathbb{V}^H(I) = \mathbb{V}^S(I) \cap \text{Proj}(R)$$

Proof Trivial. ■

Proposition 2.1.8
Let S be a commutative graded ring. Then $\text{Proj}(S)$ with the Zariski topology is a subspace of $\text{Spec}(S)$ with the Zariski topology.

Definition 2.1.9 (Projective Space over a Ring)
Let R be a commutative ring. Define the projective n space over A to be

$$\mathbb{P}^n_A = \text{Proj}(A[x_0, \dots, x_n])$$

Definition 2.1.10 (The Ideal of a Subset of Proj)
Let S be a commutative graded ring. Let $T \subseteq \text{Proj}(R)$ be a subset. Define the ideal associated to T to be

$$\mathbb{I}^H(T) = \{f \in S_+ \mid f \in P \text{ for all } P \in T\}$$

Lemma 2.1.11
Let S be a commutative graded ring. Then the following are true.

- If $T \subseteq S$ is a subset, then $\mathbb{I}^H(T)$ is a homogeneous radical ideal.
- If $T \subseteq S$ is a subset, then we have

$$\mathbb{I}^H(T) = \left(\bigcap_{P \in T} P\right) \cap S_+$$

- If $T_1 \subseteq T_2 \subseteq S$, then $\mathbb{I}^H(T_2) \subseteq \mathbb{I}^H(T_1)$.
- If $T_1, T_2 \subseteq S$, then $\mathbb{I}^H(T_1 \cup T_2) = \mathbb{I}^H(T_1) \cap \mathbb{I}^H(T_2)$.

Proposition 2.1.12 (Hilbert's Nullstellensatz for Proj) Let S be a commutative graded ring. Then the following are true.

- If $I \subseteq S_+$ is an ideal of S, then $\mathbb{I}^H(\mathbb{V}^H(I)) = \sqrt{I}_+$.
- If $T \subseteq S$ is a subset, then $\mathbb{V}^H(\mathbb{I}^H(T)) = \overline{T}$.

Proof
From Commutative Algebra 1, we have seen that $\sqrt{I}_+ = \bigcap_{I \subseteq P \text{ is relevant}} P$. Then we have

$$\mathbb{I}^H(\mathbb{V}^H(I)) = \left(\bigcap_{P \in \mathbb{V}^H(I)} P\right) \cap S_+ = \left(\bigcap_{I \subseteq P \text{ is relevant}} P\right) \cap S_+ = (\sqrt{I})_+$$

For any closed set $\mathbb{V}^H(I)$ for some homogeneous ideal $I \subseteq S$, $\mathbb{V}^H(I)$ contains T if and only if $I \subseteq P$ for all $P \in T$. And this is true if and only if

$$I \subseteq \mathbb{I}^H(T) = \left(\bigcap_{P \in T} P\right) \cap S_+$$

Since the largest homogeneous ideal contained by $\mathbb{I}^H(T)$ is itself, the smallest closed subset containing T must be $\mathbb{V}^H(\mathbb{I}^S(T))$. Hence $\overline{T} = \mathbb{V}^H(\mathbb{I}^H(T))$. ■

Proposition 2.1.13
Let S be a commutative graded ring. Then the following are true.

- There is an inclusion reversing bijection
$$\begin{Bmatrix} \text{Homogeneous radical} \\ \text{ideals } I \text{ of } S \text{ such that } I=\sqrt{I}\cap S_+ \end{Bmatrix} \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Closed Subsets of} \\ \text{Proj}(S) \end{Bmatrix}$$
induced by $\mathbb{V}^{\text{H}}(-)$ and $\mathbb{I}^{\text{H}}(-)$.
- There is an inclusion reversing bijection
$$\text{Proj}(S) \overset{1:1}{\longleftrightarrow} \begin{Bmatrix} \text{Irreducible Closed Subsets of} \\ \text{Proj}(S) \end{Bmatrix}$$
induced by $\mathbb{V}^{\text{H}}(-)$ and $\mathbb{I}^{\text{H}}(-)$.

Lemma 2.1.14
Let S be a commutative graded ring. Let $I \subseteq S_+$ be a homogeneous ideal. Then the following are equivalent.

- $\mathbb{V}^H(I) = \emptyset$.
- If $(f_i) = I$ and each f_i is homogeneous for $i \in J$, then $\text{Proj}(S) = \bigcup_{i\in J} D_+(f_i)$.
- $\sqrt{I} \supseteq S_+$.

Thus any ideal I whose radical contains S_+ is geometrically irrelevant: Indeed if $\sqrt{I} \supseteq S_+$ then $\mathbb{V}^H(I)$ is empty.

2.2 Topological Properties of Proj

Definition 2.2.1 (Projective Distinguished Open Sets)
Let S be a commutative graded ring. Let $I \subseteq S$ be a homogeneous ideal. Define the projective distinguished open set of I to be
$$D_+(I) = \text{Proj}(S) \setminus \mathbb{V}^H(I)$$
If $I = (f)$ then we call $D_+(I)$ a basic open set.

Lemma 2.2.2
Let S be a commutative graded ring. Then the basic open sets form a basis for the Zariski topology on $\text{Proj}(S)$.

Lemma 2.2.3
Let S be a commutative graded ring. Let $f \in S$ be a homogeneous element of non-zero degree. Then there is an isomorphism of locally ringed spaces
$$D_+(f) \cong \text{Spec}((S_f)_0)$$
given by the correspondence of prime ideals of localization of graded rings.

Proposition 2.2.4
Let S be a commutative graded ring. Then the following are true.

- If S is Noetherian, then $\text{Proj}(S)$ is Noetherian.
- If S is an integral domain, then $\text{Proj}(S)$ is irreducible.

Proof

-
- I claim that $\text{Proj}(S) = \overline{(0)}$. Indeed, we have $P \in \overline{\{(0)\}}$ if and only if $P \in \mathbb{V}^H(\mathbb{I}^H(\{(0)\}))$ if and only if $\mathbb{I}^H(\{(0)\}) \subseteq P$. But since $\mathbb{I}^H(\{(0)\}) = \emptyset$, the condition is null and so we have equality.

Now suppose that $\text{Proj}(S) = \mathbb{V}^H(I) \cup \mathbb{V}^H(J)$ for some homogeneous ideals I, J of S. Without loss of generality suppose that $(0) \in \mathbb{V}^H(I)$. By definition of closure, we must have $\overline{(0)} = \mathbb{V}^H(I)$ so $\text{Proj}(S)$ is irreducible.

2.3 The Structure Sheaf on a Projective Spectrum

Definition 2.3.1 (The Structure Sheaf on Projective Spectra) Let S be a graded ring. Construct $\mathcal{O}_{\text{Proj}(S)} : \textbf{Open}(\text{Proj}(S)) \to \textbf{Rings}$ as follows.

- For $U \subseteq \text{Proj}(S)$ an open set, define

$$\mathcal{O}_{\text{Proj}(S)}(U) = \left\{ (s_p)_{p \in U} \in \prod_{p \in U} S_{(p)} \;\middle|\; \begin{matrix} \forall p, \exists p \in U_p \subseteq U \text{ open and } a, f \in S \text{ homogeneous s.t.} \\ q \in U_p \text{ implies } f \notin q \text{ and } s_q = a/f \in S_{(q)} \end{matrix} \right\}$$

- For $V \subseteq U$ the inclusion, define the unique map $\mathcal{O}_{\text{Proj}(S)}(U) \to \mathcal{O}_{\text{Proj}(S)}(V)$ by projection map

$$\text{res}^U_V : \prod_{p \in U} S_{(p)} \to \prod_{p \in V} S_{(p)}$$

Then $\mathcal{O}_{\text{Proj}(S)}$ is a sheaf on S.

Proposition 2.3.2
Let S be a commutative graded ring. Then the following are true.
- For any $p \in \text{Proj}(S)$, there is a graded ring isomorphism $\mathcal{O}_{\text{Proj}(S),p} \cong (S_{(p)})_0$.
- For any homogeneous element $f \in S_+$, there is an isomorphism $\mathcal{O}_{\text{Proj}(S)}(D_+(f)) \cong S_{(f)}$

2.4 The Associated Sheaf of Graded Modules

Definition 2.4.1 (Sheaves of Modules on Graded Rings) Let S be a graded ring. Let M be a graded S-module. Consider the module

$$M_{(p)} = T^{-1}M$$

where T is the multiplicative system of homogenous elements of S not in p. Define the sheaf associated to M on $\text{Proj}(S)$,

$$\widetilde{M} : \textbf{Open}(\text{Proj}(S)) \to \textbf{Rings}$$

as follows.
- For each $U \subseteq \text{Proj}(S)$ open, define

$$\mathcal{O}_{\text{Proj}(S)}(U) = \left\{ s : U \to \prod_{p \in U} M_{(p)} \;\middle|\; \begin{matrix} \forall p \in U,\ s(p) \in M_p \text{ and } \exists\ U_p \subseteq U \text{ s.t. } q \in V \text{ implies} \\ s(q) = \frac{m}{f} \in M_q \text{ for } f \in S \text{ and } m \in M \text{ homogenous} \end{matrix} \right\}$$

- For $V \subseteq U$ an inclusion, define the unique morphism $\widetilde{M}(U) \to \widetilde{M}(V)$ by restriction.

Proposition 2.4.2 Let S be a graded ring and let M be a graded module over S. Then the following are true regarding the sheaf of modules $\widetilde{M}$.
- For any $p \in \text{Proj}(S)$, there is an isomorphism $\widetilde{M}_p \cong M_{(p)}$
- For any homogenous $f \in S_+$, there is an isomorphism

$$\widetilde{M}|_{D_+(f)} \cong \widetilde{M_{(f)}}$$

via the isomorphism of $D_+(f)$ with $\text{Spec}(S_{(f)})$

Lemma 2.4.3 Let S be a graded ring and let M be a graded module over S. Then $\widetilde{M}$ is a $\mathcal{O}_{\text{Proj}(S)}$-module. Moreover, if S is Noetherian, then $\widetilde{M}$ is a coherent $\mathcal{O}_{\text{Proj}(S)}$-module.

For a graded ring S, recall that we can shift the grading of S up and down, and it will still be a graded ring. This is the shifted $S(n)$ where n denotes shifting up n times. (Note that this is not true for algebras because S is an algebra over S_0, if the grading is shifted then $S(n)$ is an algebra over $S(n)_n$.

Definition 2.4.4 (The Twisting Sheaf of Serre) Let S be a graded ring. Let $X = \text{Proj}(S)$. For any $n \in \mathbb{Z}$, define the sheaf

$$\mathcal{O}_X(n) = \widetilde{S(n)}$$

We call $\mathcal{O}_X(1)$ the twisting sheaf of Serre. For any sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$, denote

$$\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$$

The twisting sheaf of Serre is important because it is the prototypical example of an invertible sheaf on $\text{Proj}(S)$.

Proposition 2.4.5 Let S be a graded ring and let $X = \text{Proj}(S)$. Suppose that S is generated by S_1 as an S_0-algebra. Then the following are true.

- The sheaf $\mathcal{O}_X(n)$ is invertible.
- If M is a graded S-module, then $\widetilde{M}(n) \cong \widetilde{M(n)}$
- There is an isomorphism $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$

Definition 2.4.6 (Graded Module Associated to a Sheaf of Modules) Let S be a graded ring and let $X = \text{Proj}(S)$. Let $\mathcal{F}$ be a sheaf of mO_X-module. Define the graded S-module associated to $\mathcal{F}$ to be the group

$$\Gamma_*(\mathcal{F}) = \bigoplus_{n\in\mathbb{Z}} \Gamma(X, \mathcal{F}(n))$$

together with the structure of graded S-module as follows. If $s \in S_d$, then s determines a global section $s \in \Gamma(X, \mathcal{O}_X(d))$ naturally. For any $t \in \Gamma(X, \mathcal{F}(n))$, define $s \cdot t \in \Gamma(X, \mathcal{F}(n+d))$ by sending $s \otimes t \in \mathcal{F}(n) \otimes \mathcal{O}_X(d)$ to $\mathcal{F}(n+d)$ by the isomorphism

$$\mathcal{F}(n) \otimes \mathcal{O}_X(d) \cong \mathcal{F}(n+d)$$

Lemma 2.4.7 Let A be a ring and let $S = A[x_0, \dots, x_n]$ for $r \geq 1$. Then there is an isomorphism

$$\Gamma_*(\mathcal{O}_{\text{Proj}(S)}) \cong S$$

Note that this is not true if S is not a polynomial ring.

3 The Category of Schemes

3.1 The Definition of Schemes

Definition 3.1.1 (Affine Schemes)
Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is an affine scheme if there exists a commutative ring R such that $(X, \mathcal{O}_X) \cong (\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$.

Definition 3.1.2 (The Affine Space)
Let R be a commutative ring. Let $n \in \mathbb{N} \setminus \{0\}$. Define the n dimensional affine space over R to be

$$\mathbb{A}_R^n = \text{Spec}(R[x_1, \dots, x_n])$$

Proposition 3.1.3
Let X be an affine scheme. Then X is compact.

Definition 3.1.4 (The Category of Affine Schemes)
The category of affine schemes **AffSch** is the full subcategory of **LRS** consisting of affine schemes.

Lemma 3.1.5
There is an equivalence of categories

$$\mathbf{CRing} \cong \mathbf{AffSch}^{\text{op}}$$

given by Spec whose inverse is Γ.

Definition 3.1.6 (Projective Scheme)
Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is a projective scheme if there exists a commutative graded ring S such that $(X, \mathcal{O}_X) \cong (\text{Proj}(S), \mathcal{O}_{\text{Proj}(S)})$.

Definition 3.1.7 (Schemes)
Let $(X, \mathcal{O}_X)$ be a locally ringed space. We say that X is a scheme if for all $p \in X$, there exists an open neighborhood $U \subseteq X$ of p such that $(U, \mathcal{O}_X|_U)$ is an affine scheme.

Proposition 3.1.8
Let S be a commutative graded ring. Then $(\text{Proj}(S), \mathcal{O}_{\text{Proj}(S)})$ is a scheme.

Lemma 3.1.9
Let R be a commutative ring. Let $f \in R$ be an element. Then there is a natural isomorphism of locally ringed space

$$(\text{Spec}(R_f), \mathcal{O}_{\text{Spec}(R_f)}) \cong (D(f), \mathcal{O}_{\text{Spec}(R)}|_{D(f)})$$

Proof We already established a natural homeomorphism $\varphi : D(f) \overset{\cong}{\to} \text{Spec}(R_f)$ given by the correspondence of prime ideals of localization $\varphi(P) = P_f$. It remains to define an isomorphism of sheaves. For each open set $U \subseteq \text{Spec}(R_f)$, define $f(U) : \mathcal{O}_{\text{Spec}(R_f)}(U) \to \mathcal{O}_{\text{Spec}(R)}|_{D(f)}(\varphi^{-1}(U))$ as follows. For each $p \in U$, there is a natural isomorphism $(R_f)_P \cong R_p$. Since $\mathcal{O}_{\text{Spec}(R_f)}(U)$ is a subset of $\prod_{p \in U}(R_f)_p$, define $f(U)$ by the product of the isomorphisms $(R_f)_p \cong R_p$. Evidently, $f(U)$ is an isomorphism since φ is a homeomorphism. Thus we are done. ∎

Lemma 3.1.10
Let $(X, \mathcal{O}_X)$ be a scheme. Let $U \subseteq X$ be an open set. Then $(U, \mathcal{O}_X|_U)$ is a scheme.

Proof
Let $p \in U$. Since X is a scheme, there exists an open neighborhood $V \subseteq X$ of p such that $(V, \mathcal{O}_X|_V)$ is an affine scheme. Since V has an open cover of basic open sets, there exists a basic open set $D(f)$ such that $p \in D(f)$. But by the above, $D(f)$ is an affine scheme. Hence p lies locally in an affine scheme. Thus X is a scheme. ■

Example 3.1.11
The open subscheme $\mathbb{A}^1_{\mathbb{C}} \setminus \{(t)\} \subseteq \mathbb{A}^1_{\mathbb{C}} = \text{Spec}(\mathbb{C}[t])$ is an affine scheme.

Proof
I claim that there is an isomorphism of schemes

$$(\mathbb{A}^1_{\mathbb{C}} \setminus \{(t)\}, \mathcal{O}_{\mathbb{A}^1_{\mathbb{C}} \setminus \{(t)\}}) \cong \text{Spec}\left(\frac{\mathbb{C}[x,y]}{(xy-1)}\right)$$

Firstly notice that $\text{Spec}\left(\frac{\mathbb{C}[x,y]}{(xy-1)}\right) = \{(x-c, y-1/c) \mid c \in \mathbb{C}\} \cup \{(0)\}$.

Define a map of sets $\mathbb{A}^1_{\mathbb{C}} \setminus \{(t)\} \to \text{Spec}\left(\frac{\mathbb{C}[x,y]}{(xy-1)}\right)$ by $(t-c) \mapsto (x-c, y-1/c)$ and $(0) \mapsto (0)$. Clearly it is bijective. Now notice that any proper closed set of the two spaces consists of a finite number of closed points. Hence the map of sets is a homeomorphism. ■

3.2 Understanding Schemes Locally

Theorem 3.2.1 (Gluing Schemes)
Let $(X_i, \mathcal{O}_{X_i})$ for $i \in I$ be a family of schemes. Suppose that we have the following data.

- For each $i, j \in I$, an open subset $U_{ij} \subseteq X_i$.
- Isomorphisms $\theta_{ij} : U_{ij} \xrightarrow{\cong} U_{ji}$.

such that the following are true.

- $\theta_{ii} = \text{id}$.
- $\theta_{ij}|_{U_{ij} \cap U_{ik}} \circ \theta_{jk}|_{U_{ji} \cap U_{jk}} = \theta_{ik}|_{U_{ij} \cap U_{ik}}$.

Then there exists a unique scheme $(X, \mathcal{O}_X)$ and an open cover $X = \bigcup_{i \in I} X'_i$ and a family of isomorphisms $\varphi_i : (X'_i, \mathcal{O}_X|_{X'_i}) \to (X_i, \mathcal{O}_{X_i})$ such that

$$(\varphi_j|_{X_i \cap X_j})^{-1} \circ \theta_{ij} \circ \varphi_i|_{X_i \cap X_j} = \text{id}$$

for all $i, j \in I$.

Proposition 3.2.2 (Gluing Morphisms)
Let X, Y be a scheme. Let $X = \bigcup_{i \in I} U_i$ be an open cover of X. Let $f_i : U_i \to Y$ be morphisms of schemes such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all $i, j \in I$. Then there exists a unique morphism of schemes $f : X \to Y$ such that $f|_{U_i} = f_i$ for all $i \in I$.

Proof
Since morphisms of topological spaces and morphisms of sheaves glue uniquely, we are done.

We note that the unique morphism of schemes is defined as follows. The underlying map of topological spaces is given by $f(p) = f_i(p)$ for $p \in U_i$. For the morphism of sheaves $\mathcal{O}_Y \to f_*\mathcal{O}_X$, let $V \subseteq Y$ be an open subset. Then define $\mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ as follows. For each $s \in \mathcal{O}_Y(V)$, the morphisms $(f_i)^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_{U_i}((f_i)^{-1}(V))$ gives a section $(f_i)^\#(V)(s) \in \mathcal{O}_{U_i}((f^{-1}(V)) = \mathcal{O}_X(U_i \cap f^{-1}(V))$. Then the identity and gluing axiom guarantees a unique section $f(s) \in \mathcal{O}_X(f^{-1}(V))$. ■

Example 3.2.3
Let k be an algebraically closed field. Let $U = \mathbb{A}^2_k \setminus \{(x,y)\}$. Then the following are true.

- $\Gamma(U) \cong k[x, y]$.
- U is not an affine scheme.

Definition 3.2.4 (The Affine Communication Property)
Let X be a scheme. Let P be a property of affine open subsets. We say that P has the affine communication property if the following are true.
- If $\text{Spec}(R) \subseteq X$ has property P, then $\text{Spec}(R_f) \subseteq X$ has property P for any $f \in R$.
- If $\text{Spec}(R) \subseteq X$, $R = (f_1, \dots, f_n)$ and $\text{Spec}(R_{f_i}) \subseteq X$ has property P for all i, then $\text{Spec}(R) \subseteq X$ has property P.

Theorem 3.2.5 (The Affine Communication Lemma)
Let X be a scheme. Let P be a property of affine open subsets of a scheme such that P has the affine communication property. Let $X = \bigcup_{i \in I} \text{Spec}(R_i)$ is an affine open cover such that each $\text{Spec}(R_i)$ has property P, then every affine open subset of X has property P.

3.3 Properties of the Category of Schemes

Definition 3.3.1 (The Category of Schemes)
The category **Sch** of schemes is the full subcategory of **LRS** consisting of schemes.

Explicitly, a morphism of schemes $f : X \to Y$ consists of the following data.

- A continuous map $X \to Y$.
- A morphism of sheaves $\mathcal{O}_Y \to f_* \mathcal{O}_X$.

Proposition 3.3.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism of ringed spaces. Then the following are equivalent.
- f is a morphism of schemes.
- For any affine open subsets $U \subseteq X$ and $V \subseteq Y$ such that $f(U) \subseteq V$, the morphism of ringed spaces $f|_U : U \to V$ is a morphism of the form induced by the Spec functor.
- There exists affine open covers $X = \bigcup_{i \in I} U_i$ and $Y = \bigcup_{i \in I} V_i$ such that $f|_{U_i}$ is a morphism of the form induced by the Spec functor.

Proof
$(1) \implies (2)$: Since f is a morphism of locally ringed spaces, $f|_U : U \to V$ is also a morphism of locally ringed spaces (stalks of f are isomorphic to stalks of $f|_U$). Since $\text{Spec} : \mathbf{CRing} \to \mathbf{LRS}$ is fully faithful, $f|_U$ must be induced by the Spec functor.

$(2) \implies (3)$: Trivial.

$(3) \implies (1)$: Since stalks of f are equal to the stalks of $f|_{U_i}$ and morphisms induced by the spec functor are morphisms of locally ringed spaces, f must also be a morphism of locally ringed spaces. ■

Proposition 3.3.3
There is an adjunction

$$\Gamma : \mathbf{Sch} \overset{\dashv}{\rightleftarrows} \mathbf{CRing} : \text{Spec}$$

Proof
We wish to show that there is a natural isomorphism

$$\text{Hom}_{\mathbf{Sch}}(X, \text{Spec}(R)) \cong \text{Hom}_{\mathbf{CRing}}(R, \Gamma(X))$$

We first construct the canonical morphism $X \to \text{Spec}(\Gamma(X))$. Let $X = \bigcup_{i \in I} U_i$ be an affine open cover. For each $i \in I$, define $f_i : \text{Spec}(\mathcal{O}_X(U_i)) \cong U_i \to \text{Spec}(\Gamma(X))$ to be the morphism $\text{Spec}(\Gamma(X) = \mathcal{O}_X(X) \overset{\text{res}^X_{U_i}}{\to} \mathcal{O}_X(U_i))$. Since $\text{res}^X_{U_i} \circ \text{res}^X_{U_i \cap U_j} = \text{res}^X_{U_j} \circ \text{res}^X_{U_i \cap U_j}$, by 3.2.2 there exists a unique morphism $f : X \to \text{Spec}(\Gamma(X))$ such that $f|_{U_i} = \text{Spec}(\text{res}^X_{U_i})$. By construction, the global section of the morphism of sheaves $\Gamma(f) : \Gamma(X) \to \Gamma(\text{Spec}(\Gamma(X))) \cong \Gamma(X)$ is the identity map. Moreover, notice that if X is affine, the the morphism $X \to \text{Spec}(\Gamma(X))$ is an isomorphism.

Now I claim that for any morphism $g : X \to \text{Spec}(R)$, there exists a unique ring homomorphism $R \to \Gamma(X)$ such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[rr, "g"] \arrow[dr, "f"'] & & \text{Spec}(R) \\ & \text{Spec}(\Gamma(X)) \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

and thus completing the proof (surjectivity given by composition with $X \to \text{Spec}(\Gamma(X))$. Since the morphism $X \to \text{Spec}(\Gamma(X))$ is natural and morphisms glue, it suffices to check for the case that X is affine. But if X is affine, then f is an isomorphism hence we are done. ■

Corollary 3.3.4
Let R be a commutative ring. Then $\text{Spec}(R)$ is the terminal object of $\mathbf{Sch}_R$.

Proof Let X be a scheme over R. By the above, there is a natural isomorphism

$$\text{Hom}_{\mathbf{Sch}}(X, \text{Spec}(R)) \cong \text{Hom}_{\mathbf{CRing}}(R, \Gamma(X)) \cong \text{Hom}_{\mathbf{Alg}_R}(R, \Gamma(X))$$

Since R is the initial object of $\mathbf{Alg}_R$, the hom set consists of one element. Hence $\text{Spec}(R)$ is the terminal object of $\mathbf{Sch}_R$. ■

Proposition 3.3.5 Let X and Y be two schemes over S. Then the fibered product $X \times_S Y$ exists and is unique up to unique isomorphism.

Proof We first prove the theorem for the case of affine schemes. Let $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$ and $S = \text{Spec}(C)$. I claim that $\text{Spec}(A \otimes_C B)$ is the fibered product of X and Y over S. Using the equivalence of categories, we have that

$$\begin{aligned} \text{Hom}_{\mathbf{AffSch}}(Z, \text{Spec}(A \otimes_C B)) &\cong \text{Hom}_{\mathbf{Rings}}(A \otimes_C B, \Gamma(Z)) \\ &\cong \text{Hom}_{\mathbf{Rings}}(A, \Gamma(Z)) \times_{\text{Hom}_{\mathbf{Rings}}(C, \Gamma(Z))} \text{Hom}_{\mathbf{Rings}}(B, \Gamma(Z)) \\ &\cong \text{Hom}_{\mathbf{AffSch}}(Z, \text{Spec}(A)) \times_{\text{Hom}_{\mathbf{AffSch}}(Z, \text{Spec}(C))} \text{Hom}_{\mathbf{AffSch}}(Z, \text{Spec}(B)) \end{aligned}$$

Thus we have proved that $\text{Spec}(A \otimes_C B)$ is the fiber product of X and Y over S. ■

3.4 More on Projective Schemes

Lemma 3.4.1
Let R be a commutative ring. Then $\mathbb{P}^n_R$ is isomorphic to the scheme defined by the following gluing data.

- For $0 \leq i \leq n$, the affine schemes $U_i = \text{Spec}(R[x_0/x_i, \dots, \widehat{x_i/x_i}, \dots, x_n/x_i])$.
- For each $0 \leq i \neq j \leq n$, define

$$\varphi_{ij} : D(x_j/x_i) \subseteq U_i \to D(x_i/x_j) \subseteq U_j$$

by the isomorphism $R[x_0/x_j, \dots, \widehat{x_j/x_j}, \dots, x_n/x_j]_{x_i/x_j} \to R[x_0/x_i, \dots, \widehat{x_i/x_i}, \dots, x_n/x_i]_{x_j/x_i}$ that is multiplication by x_j/x_i, considered as subrings of $R[x_0, \dots, x_n, x_0^{-1}, \dots, x_n^{-1}]$.

Proposition 3.4.2
Let $S = \bigoplus_{i=0}^\infty S_i$ be a commutative graded ring. Suppose that S is a finitely generated S_0-algebra. Then there exists $d \in \mathbb{N}$ such that $\bigoplus_{i=0}^\infty S_{di}$ is a finitely generated S_0-module with generators in degree 1.

Proposition 3.4.3
Let $S = \bigoplus_{i=0}^\infty S_i$ be a commutative graded ring. Then there is an isomorphism

$$\text{Proj}(S) \cong \text{Proj}\left(\bigoplus_{i=0}^\infty S_{ni}\right)$$

for any $n \in \mathbb{N} \setminus \{0\}$.

Corollary 3.4.4
Let $S = \bigoplus_{i=0}^\infty S_i$ be a commutative graded ring. Suppose that S is a finitely generated S_0-algebra. Then there exists a commutative graded ring $T = \bigoplus_{i=0}^\infty T_i$ that is a finitely generated T_0-module with generators in degree 1 such that

$$\text{Proj}(S) \cong \text{Proj}(T)$$

Definition 3.4.5 (Projective Space over a Scheme)
Let X be a scheme. Let $n \in \mathbb{N} \setminus \{0\}$. Define the projective n-space over X to be

$$\mathbb{P}_X^n = \mathbb{P}_\mathbb{Z}^n \times_\mathbb{Z} X$$

3.5 Interpreting Elements of the Sheaf as Functions

Definition 3.5.1 (The Residue Field)
Let X be a scheme. Let $p \in X$. Define the residue field of p to be

$$k(p) = \frac{\mathcal{O}_{X,p}}{m}$$

where m is the unique maximal ideal of $\mathcal{O}_{X,p}$.

Definition 3.5.2 (Values of Elements of a Sheaf)
Let $(X, \mathcal{O}_X)$ be a scheme. Let $U \subseteq X$ be an open set. Let $f \in \mathcal{O}_X(U)$. Let $p \in U$. Define the value of f at p to be the image of f in the morphism

$$\mathcal{O}_X(U) \to \mathcal{O}_{X,p} \to k(p)$$

We denote the value by $f(p)$.

Lemma 3.5.3
Let $(X, \mathcal{O}_X)$ be a scheme. Let $U \subseteq X$ be an open set. Let $f \in \mathcal{O}_X(U)$. Then the set

$$S = \{p \in U \mid \text{ the value of } f \text{ at } p \text{ is non-zero } \}$$

is an open subset of X.

Proof
Suppose that p lies in the given set. This means that $[(f, U)] \in \mathcal{O}_{X,p}$ is invertible. So there exists $[(g, V)] \in \mathcal{O}_{X,p}$ such that $[(f, U)] \cdot [(g, V)] = [(1, X)]$. So there exists an open neighborhood $W_p \subseteq U \cap V$ of p such that $f|_{W_p} \cdot g|_{W_p} = 1$. Notice that for any $q \in W_p$, the same equation holds in $\mathcal{O}_{X,q}$. Hence $S = \bigcup_{p \in S} W_p$ is an open set. ■

Definition 3.5.4 (Support of a Section)
Let $(X, \mathcal{O}_X)$ be a scheme. Let $f \in \mathcal{O}_X(X)$ be a global section. Define the support of f to be

$$\text{Supp}(f) = \{p \in X \mid [(f, X)] \neq 0 \in \mathcal{O}_{X,p}\}$$

Lemma 3.5.5
Let $(X, \mathcal{O}_X)$ be a scheme. Let $f \in \mathcal{O}_X(X)$ be a global section. Then $\text{Supp}(f) \subseteq X$ is a closed subset.

Proof
For any $p \notin \text{Supp}(f)$, we have that $[(f, X)] \in \mathcal{O}_{X,p}$ is zero. So there exists a neighborhood $V_p \subseteq X$ of p such that $f|_{V_p} = 0$. Then for all $q \in V_p$, clearly $[(f, X)] = 0 \in \mathcal{O}_{X,q}$. Then clearly $U = \bigcup_{p \in X} V_p$ is an open subset of X that is the complement of $\text{Supp}(f)$. Hence $\text{Supp}(f)$ is a closed subset. ∎

Notice that $\{p \in X \mid f(p) = 0\}$ is an open subset, while the support of f is closed. They are two different notions.

4 Topological Properties of Schemes

4.1 Reduced Schemes

In a general scheme, functions are not determined by points.

Example 4.1.1
Consider the spectra $X = \text{Spec}\left(\frac{k[\varepsilon]}{(\varepsilon^2)}\right)$. Then for all $p \in X$, the value of ε at p is 0.

Proof
Notice that the residue field of X at (ε) is given by

$$k((\varepsilon)) = \frac{(k[\varepsilon]/(\varepsilon^2))_{(\varepsilon)}}{(\varepsilon} \cong \frac{k[\varepsilon]/(\varepsilon^2)}{(\varepsilon)} \cong k$$

and the k-algebra homomorphism $\mathcal{O}_X(X) = \frac{k[\varepsilon]}{(\varepsilon^2)} \to k = k((\varepsilon))$ is given by $\varepsilon \to 0$. Hence the value of ε at the only point (ε) in X is 0. ■

So it is not possible to identify the functions ε and 0 only by their value at the points of the spectrum. Notice that checking whether a particular element is 0 on all points is the same as checking whether the element lies in the nilradical of the ring, because it is the intersection of prime ideals.

Definition 4.1.2 (Reduced Schemes)
Let X be a scheme. We say that X is reduced for every open set $U \subseteq X$, $\mathcal{O}_X(U)$ is a reduced ring.

Lemma 4.1.3
Let X be a scheme. Then X is reduced if and only if for all $p \in X$, $\mathcal{O}_{X,p}$ is a reduced ring.

Proof
Suppose first that X is reduced. Let $p \in X$. Let $[(f, U)] \in \mathcal{O}_{X,p}$ be nilpotent. Then $[(f^n, U)] = 0 = [(0, U)]$ for some $n \in \mathbb{N}$. By definition of the equivalence classes, we must have an open subset $V \subseteq U$ such that $(f|_V)^n = (f^n)|_V = 0$ in $\mathcal{O}_X(V)$. Since $\mathcal{O}_X(V)$ is nilpotent, we must have that $f|_V = 0$. Hence $[(f, U)] = [(f|_V, V)] = [(0|_V, V)] = 0 \in \mathcal{O}_{X,p}$. Hence $\mathcal{O}_{X,p}$ is a reduced ring.

Now suppose that for all $p \in X$, $\mathcal{O}_{X,p}$ is reduced. Let $f \in \mathcal{O}_X(U)$ be a nilpotent element. For each $p \in U$, $[(f, U)]$ is an element of $\mathcal{O}_{X,p}$. At the same time, $f^n = 0$ in $\mathcal{O}_X(U)$ implies that $[(f, U)]^n = [(f^n, U)] = 0 \in \mathcal{O}_{X,p}$ for all $p \in U$. Since $\mathcal{O}_{X,p}$ is reduced, there exists a neighborhood $V_p \subseteq U$ of p such that $f|_{V_p} = 0$. Now $\{V_p \mid p \in U\}$ is an open cover of U, and the collection $\{f|_{V_p} \mid p \in U\}$ agrees on intersections in the open cover. By the gluing axiom and the identity axiom the unique element in $\mathcal{O}_X(U)$ that restricts to $f|_{V_p} = 0$ for each $p \in U$ must be f, and so $f = 0$. Hence $\mathcal{O}_X(U)$ is reduced. ■

Lemma 4.1.4
Let R be a commutative ring. Then R is reduced if and only if Spec(R) is reduced.

Proof
Suppose that R is reduced. Let $p \in \text{Spec}(R)$. Then there is a ring isomorphism $\mathcal{O}_{\text{Spec}(R),p} \cong R_p$. Since the localization of a reduced ring is reduced, we must have that R_p is reduced. Hence by lmm 4.1.3 Spec(R) is reduced.

Conversely, Spec(R) is a reduced ring, then the ring isomorphism $\mathcal{O}_{\text{Spec}(R)}(\text{Spec}(R)) \cong R$ shows that R is reduced. ■

Lemma 4.1.5
Let X be a reduced scheme. Let $f, g \in \mathcal{O}_X(X)$. Then $f = g$ if and only if for all $p \in X$, the value of f at p is equal to the value of g at p.

Proof
If $f = g$ then the values must be the same. Now suppose that the values of f and g agree on all points. Without loss of assumption we can assume that $g = 0$ and f takes the value of 0 at all points. ∎

Definition 4.1.6 (The Reduced Scheme)
Let $(X, \mathcal{O}_X)$ be a scheme. Define the reduced scheme $(\mathcal{O}_X)_{\text{red}}$ of X to be the sheafification of the presheaf defined as follows.

- For each open set $U \subseteq X$, define $(\mathcal{O}_X)_{\text{red}}(U) = \frac{\mathcal{O}_X(U)}{N(\mathcal{O}_X(U))}$.
- For an inclusion of open sets $V \subseteq U$, define the induced map
$$(\mathcal{O}_X)_{\text{red}}(U) \to (\mathcal{O}_X)_{\text{red}}(V)$$
as follows. There is a restriction map $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$. Projecting to the quotient, we have the map $\mathcal{O}_X(U) \to (\mathcal{O}_X)_{\text{red}}(V)$. Since nilpotents are sent to nilpotents under ring homomorphisms, the map descends to the quotient $(\mathcal{O}_X)_{\text{red}}(U) \to (\mathcal{O}_X)_{\text{red}}(V)$.

Definition 4.1.7 (The Canonical Map of the Reduced Scheme)
Let X be a scheme. Define the canonical map $X_{\text{red}} \to X$ as follows.

- The map of the underlying topological space is given by the identity map.
- The morphism of sheaves $\mathcal{O}_X \to \text{id}_*((\mathcal{O}_X)_{\text{red}}) = (\mathcal{O}_X)_{\text{red}}$ is given by the quotient map $\mathcal{O}_X(U) \to (\mathcal{O}_X)_{\text{red}}(U) = \frac{\mathcal{O}_X(U)}{N(\mathcal{O}_X(U))}$ for any open set U.

Proposition 4.1.8
Let X be a scheme. Then the reduced scheme of X satisfies the following universal property. For any reduced scheme Y and morphism $f : Y \to X$, there exists a unique morphism $f_{\text{red}} : Y \to X_{\text{red}}$ such that the following diagram commutes:

$$\begin{tikzcd}
Y \arrow[r, dashed, "\exists! f_{\text{red}}"] \arrow[dr, "f"'] & X_{\text{red}} \arrow[d] \\
& X
\end{tikzcd}$$

Moreover, X_{red} is the unique reduced scheme satisfying the above.

4.2 Integral Schemes

Definition 4.2.1 (Integral Schemes)
Let X be a scheme. We say that X is integral if for every open set $U \subseteq X$, $\mathcal{O}_X(U)$ is an integral domain.

Lemma 4.2.2
Let X be a scheme. If X is integral, then $\mathcal{O}_{X,p}$ is an integral domain for all $p \in X$.

Proposition 4.2.3
Let X be a scheme. Then X is integral if and only if X is reduced and irreducible.

Proof
Suppose that X is integral. Then clearly X is reduced. If X is not irreducible, then there exists proper closed subsets $X_1, X_2 \subseteq X$ such that $X = X_1 \cup X_2$. Let $U_i = X \setminus X_i$. Then we have
$$\emptyset = X \setminus (X_1 \cup X_2) = U_1 \cap U_2$$

and U_1, U_2 are non-empty. Then we have

$$\mathcal{O}_X(U_1 \amalg U_2) \cong \mathcal{O}(U_1) \times \mathcal{O}(U_2)$$

so that $\mathcal{O}_X(U_1 \amalg U_2)$ is not an integral domain, a contradiction.

Now suppose that X is reduced and irreducible. Let $U \subseteq X$ be an open set. Let $V \subseteq U$ be an affine open subset of U. Since X is irreducible, V is also irreducible and since V is affine, by 1.2.6 we must have that $\mathcal{O}_X(V)$ is an integral domain. Now consider the morphism $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$. I claim that the morphism is injective. Let $f \in \ker(\mathcal{O}_X(U) \to \mathcal{O}_X(V))$. Then $f|_V = 0$. Notice that this is true for all $W \subseteq U$ affine open subsets. Since U is a scheme, such affine open subsets cover U, hence by the gluing axiom, $f = 0|_U$. Hence $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is injective. Now since $\mathcal{O}_X(V)$ is an integral domain, $\mathcal{O}_X(U)$ is also an integral domain since it is the subring of an integral domain. Hence X is integral. ■

Lemma 4.2.4
Let R be a commutative ring. Then the following are true.
- R is an integral domain if and only if Spec(R) is integral.
- If R is a graded integral domain, then Proj(R) is integral.

Proof
If R is an integral domain, then by 1.2.6, Spec(R) is irreducible. Since R is an integral domain, R is reduced and by 4.1.4, Spec(R) is reduced. Conversely, if Spec(R) is integral, then it is irreducible and by 1.2.6 R must be an integral domain. ■

Proposition 4.2.5
Let k be an algebraically closed field. Let X, Y be integral schemes over k. Then $X \times_k Y$ is integral.

4.3 Noetherian Schemes

Definition 4.3.1 (Locally Noetherian Schemes)
Let X be a scheme. We say that X is locally Noetherian if there exists an open affine cover $\{U_i = \text{Spec}(A_i) \mid i \in I\}$ of X such that each A_i is Noetherian.

Proposition 4.3.2
Let X be a scheme. Then X is locally Noetherian if and only if for every affine open subset $U \subseteq X$, we have $\mathcal{O}_X(U)$ is Noetherian.

Definition 4.3.3 (Noetherian Schemes)
Let X be a scheme. We say that X is Noetherian if X is locally Noetherian and compact.

Lemma 4.3.4
Let X be a scheme. Suppose that X is Noetherian. Then the following are true.
- The underlying space of X is Noetherian.
- Every open subset of X is compact.

Proof
Since X is Noetherian, there exists Noetherian rings $R_1, \dots, R_n$ such that $X = \bigcup_{i=1}^n \text{Spec}(R_i)$. By 1.2.6, each Spec$(R_i)$ is Noetherian. We have seen that a finite union of Noetherian subspaces is Noetherian in Algebraic Geometry 1. Hence the underlying space of X is Noetherian.

We have also seen in Algebraic Geometry 1 that every open subset of a Noetherian space is compact. ■

Lemma 4.3.5
Let R be a commutative ring. Then R is Noetherian if and only if Spec(R) is a Noetherian scheme.

If R is Noetherian, then the underlying topological space of Spec(R) is also Noetherian. But the converse of this is not true in general.

4.4 Dimension and Codimension

Definition 4.4.1 (The Dimension of a Scheme)
Let X be a scheme. Define the dimension of X to be dimension of the underlying space of X. Explicitly, we have

$$\dim(X) = \sup_{\substack{Z_0,\dots,Z_n\subseteq X \\ \text{closed irreducible subsets}}} \{n \in \mathbb{N} \mid Z_0 \subset Z_1 \subset \cdots \subset Z_n\}$$

Proposition 4.4.2
Let X be a scheme. Then the following numbers are equal.

- $\dim(X)$.
- $\max\{\dim(V_i) \mid 1 \leq i \leq n\}$ where $V_1, \dots, V_n$ are the irreducible components of X.
- $\sup\{\dim(\mathcal{O}_{X,p}) \mid p \in X\}$.

Proposition 4.4.3
Let X, Y be schemes. Let $i : Y \to X$ be a closed immersion. If X is integral and $\dim(X) = \dim(Y)$, then i is an isomorphism.

Proposition 4.4.4
Let k be a field. Let X be an irreducible scheme of finite type over k. Then all maximal chains of closed irreducible subsets have length $\dim(X)$.

Proposition 4.4.5
Let k be a field. Let X, Y be schemes locally of finite type over k. Then we have

$$\dim(X \times_k Y) = \dim(X) + \dim(Y)$$

Definition 4.4.6 (The Codimension of a Scheme)
Let X be a scheme. Let $Y \subseteq X$ be a closed subset of X. Define the codimension of Y in X to be

$$\text{codim}_X(Y) = \sup_{\substack{Z\subseteq Y \text{ is a} \\ \text{closed irreducible subsets}}} \{n \in \mathbb{N} \mid Z = Z_0 \subset \cdots \subset Z_n \subset X \text{ are closed irreducible subsets }\}$$

Proposition 4.4.7
Let X be a scheme. Let $Y \subseteq X$ be an irreducible closed subset. Let η be the generic point of Y. Then

$$\text{codim}_X(Y) = \dim(\mathcal{O}_{X,\eta})$$

Proposition 4.4.8
Let k be a field. Let X be an irreducible scheme of finite type over k. Let Y be an irreducible closed subset of X. Then we have

$$\dim(Y) + \text{codim}_X(Y) = \dim(X)$$

4.5 Geometric Properties

Definition 4.5.1 (Geometrically Reduced Schemes)
Let k be a field. Let X be a scheme over k. We say that X is geometrically reduced if $X \times_k \overline{k}$ is reduced.

Proposition 4.5.2
Let k be a field. Let X be a scheme over k. Then the following are true.
- X is geometrically reduced.
- $X \times_k k_p$ is reduced, where k_p is the perfect closure of k.
- For all field extensions K/k, $X \times_k K$ is reduced.

Definition 4.5.3 (Geometrically Connected)

5 Sheaf of Modules over a Scheme

5.1 Sheaves on Distinguished Affine Base

Definition 5.1.1 (The Distinguished Affine Base)
Let X be a scheme. Define the distinguished affine base $\mathbf{Open}_D(X)$ of X to be the full subcategory of $\mathbf{Open}(X)$ consisting of affine open sets of the form $\text{Spec}(R)$ and $\text{Spec}(R_f)$ for all $f \in R$.

Lemma 5.1.2
Let X be a scheme. Then $\mathbf{Open}_D(X)$ is a filtered subcategory of $\mathbf{Open}(X)$.

Recall from Sheaf Theory that sheaves are uniquely defined on filtered subcategories of the category of open sets.

Corollary 5.1.3
Let X be a space. Let $\mathcal{C}$ be an algebraic category. Then there is an equivalence of categories

$$\mathcal{C}(X) = \mathcal{C}_{\mathbf{Open}_D(X)}(X)$$

Thus it suffices to define the values of a sheaf or a morphism of sheaves only on a distinguished base of X.

5.2 Quasi-Coherent Modules

Proposition 5.2.1 Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a $\mathcal{O}_X$-module. Then the following are equivalent.

- $\mathcal{F}$ is quasi-coherent.
- For all affine open subschemes U, $\mathcal{F}|_U \cong \widetilde{M}$ for some $\mathcal{O}_X(U)$-module M.

Proposition 5.2.2
Let X be a scheme. Let $\mathcal{F} \to \mathcal{G} \to \mathcal{H}$ be an sequence of of quasi-coherent sheaves on X. Then the sequence is exact if and only if for any affine open subset $U \subseteq X$, we have

$$\mathcal{F}(V) \to \mathcal{G}(V) \to \mathcal{H}(V)$$

is an exact sequence of $\mathcal{O}_X(V)$-modules for any $V \subseteq U$ open.

Proposition 5.2.3
Let $(X, \mathcal{O}_X)$ be a scheme. Then $\mathbf{QCoh}_{\mathcal{O}_X}$ is an abelian category.

Proposition 5.2.4
Let $(X, \mathcal{O}_X)$ be scheme. Let $\mathcal{F}$ be sheaf of $\mathcal{O}_X$-modules. Then $\mathcal{F}$ is quasi-coherent if and only if for all affine open subsets $\text{Spec}(R) = U \subseteq X$ and $f \in R$, the unique morphism

$$\begin{tikzcd}
\mathcal{F}(\text{Spec}(R)) \arrow[rr, "\text{Spec}(\iota)"] \arrow[dr, "-\otimes_R R_f"'] & & \mathcal{F}(\text{Spec}(R_f)) \\
& \mathcal{F}(\text{Spec}(R))_f \arrow[ur, dashed, "\exists!"'] &
\end{tikzcd}$$

given by the universal property of localization is an isomorphism.

5.3 Sheaves of Finite Type and Finitely Presented

Proposition 5.3.1
Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then the following are equivalent.
- $\mathcal{F}$ is a sheaf of finite type
- For all affine open subschemes U, $\mathcal{F}(U)$ is a finitely generated $\mathcal{O}_X(U)$-module.
- There exists an affine open cover $\{U_i \mid i \in I\}$ of X such that $\mathcal{F}(U_i)$ is a finitely generated $\mathcal{O}_X(U_i)$-module.

Definition 5.3.2 (Finitely Presented Sheaves)
Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. We say that $\mathcal{F}$ is a finitely presented sheaf if for all affine open subschemes U, $\mathcal{F}(U)$ is a finitely presented $\mathcal{O}_X(U)$-module.

Proposition 5.3.3
Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then the following are equivalent.
- $\mathcal{F}$ is finitely presented.
- There exists an affine open cover $\{U_i \mid i \in I\}$ of X such that $\mathcal{F}(U_i)$ is a finitely presented $\mathcal{O}_X(U_i)$-module.

Lemma 5.3.4
Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. If $\mathcal{F}$ is finitely presented, then $\mathcal{F}$ is finitely generated.

5.4 Coherent Modules

Proposition 5.4.1
Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then the following are equivalent.
- $\mathcal{F}$ is a coherent sheaf
- For all affine open subschemes U, $\mathcal{F}(U)$ is a coherent $\mathcal{O}_X(U)$-module.
- There exists an affine open cover $\{U_i \mid i \in I\}$ of X such that $\mathcal{F}(U_i)$ is a coherent $\mathcal{O}_X(U_i)$-module.

Lemma 5.4.2
Let $(X, \mathcal{O}_X)$ be a scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. If $\mathcal{F}$ is coherent, then $\mathcal{F}$ is finitely presented.

Proposition 5.4.3
Let $(X, \mathcal{O}_X)$ be a locally Noetherian scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then $\mathcal{F}$ is a coherent sheaf if and only if $\mathcal{F}$ is quasi-coherent and the $\mathcal{O}_X(U)$-modules M for $\mathcal{F}|_U \cong \widetilde{M}$ for U an affine open subset can be chosen so that they are finitely generated.

Proposition 5.4.4
Let $(X, \mathcal{O}_X)$ be a locally Noetherian scheme. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then the following are equivalent.
- $\mathcal{F}$ is of finite type.
- $\mathcal{F}$ is finitely presented.
- $\mathcal{F}$ is a coherent sheaf.

5.5 Adjunction Formulas

Proposition 5.5.1 Let R be a commutative ring. Then there is an equivalence of categories

$$\mathbf{Mod}_R \cong \mathbf{QCoh}_{\mathcal{O}_{\text{Spec}(R)}}$$

given by $\widetilde{(-)}$ and Γ.

Proposition 5.5.2 Let R be a Noetherian commutative ring. Then there is an equivalence of categories

$$\mathbf{Coh}_R \cong \mathbf{Coh}_{\mathcal{O}_{\text{Spec}(R)}}$$

given by $\widetilde{(-)}$ and Γ.

Proposition 5.5.3
Let S be a graded ring. There is an adjunction given by

$$\widetilde{(-)} : \mathbf{GrMod}_S \rightleftarrows \mathbf{QCoh}_{\mathcal{O}_{\text{Proj}(S)}} : \Gamma_\bullet$$

Explicitly, there is a natural isomrophism

$$\text{Hom}_{\mathbf{QCoh}_{\mathcal{O}_{\text{Proj}(S)}}}(\widetilde{M}, \mathcal{F}) \cong \text{Hom}_{\mathbf{GrMod}_S}(M, \Gamma_\bullet(\mathcal{F}))$$

for M a graded S-module and $\mathcal{F}$ a $\mathcal{O}_{\text{Proj}(S)}$-module.

We summarize all the adjunction formulas and equivalence of categories in the following list.

- There is an adjunction

$$\Gamma : \mathbf{Sch} \overset{\dashv}{\rightleftarrows} \mathbf{CRing} : \text{Spec}$$

  It restricts to an equivalence of categories

$$\mathbf{Sch} \cong \mathbf{CRing}$$

- Let R be a commutative ring. Then there is an adjunction

$$\widetilde{(-)} : \mathbf{Mod}_R \overset{\dashv}{\rightleftarrows} \mathbf{Mod}_{\mathcal{O}_{\text{Spec}(R)}} : \Gamma$$

  It restricts to equivalence of categories

$$\mathbf{Mod}_R \cong \mathbf{QCoh}_{\mathcal{O}_{\text{Spec}(R)}} \quad \text{and} \quad \mathbf{Coh}_R \cong \mathbf{Coh}_{\mathcal{O}_{\text{Spec}(R)}}$$

- Let S be a commutative graded ring. Then there is an adjunction

$$\widetilde{(-)} : \mathbf{GrMod}_S \rightleftarrows \mathbf{QCoh}_{\mathcal{O}_{\text{Proj}(S)}} : \Gamma_\bullet$$

5.6 Connections to Associated Primes

We recall some notions from Commutative Algebra 1. Let R be a commutative ring. Let M be an R-module.

- The support of M is

$$\text{Supp}(M) = \{P \in \text{Spec}(R) \mid M_P \neq 0\}$$

We recall the following facts:

- If M is finitely generated, then

$$\text{Supp}(M) = \mathbb{V}^{\mathrm{S}}(\text{Ann}_R(M))$$

- If R is Noetherian, then

$$\bigcup_{P \in \text{Ass}(M)} P = \{r \in R \mid r \text{ is a zero divisor of } M\} \cup \{0\}$$

Recall from Commutative Algebra 1 6.2.3 that i

Proposition 5.6.1
Let R be a Noetherian commutative ring. Let M be a finitely generated R-module. Then

$$\text{Supp}(M) = \overline{\text{Ass}_R(M)}$$

5.7 Vector Bundles over a Scheme

6 Points on A Scheme

6.1 The Functor of Points

Let k be a field. Recall the Yoneda Embedding to be the functor

$$\mathbf{Sch}_k \to \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}}}(\mathbf{Sch}_k, \mathbf{Set})$$

given by the following data.

- A scheme X over k is assigned to the functor $\mathrm{Hom}_{\mathbf{Sch}_k}(-, X) : \mathbf{Sch}_k \to \mathbf{Set}$.
- A morphism $f : X \to Y$ over k is assigned to the natural transformation $\mathrm{Hom}_{\mathbf{Sch}_k}(-, Y) \to \mathrm{Hom}_{\mathbf{Sch}_k}(-, X)$ given by precomposition with f.

The Yoneda lemma then says that this is a fully faithful functor. One can restrict the Yoneda embedding to the category of affine schemes, and using the equivalence of the category of finitely generated schemes of finite type over k with the category of finitely generated k-algebras, we obtain the following series of definitions.

Definition 6.1.1 (R-Points of a Scheme)
Let S be a scheme. Let X, Z be schemes over S. Define the Z-points of X to be

$$X(Z) = \mathrm{Hom}_{\mathbf{Sch}_S}(Z, X)$$

Definition 6.1.2 (The Functor of Points)
Let S be a scheme. Let X, Z be schemes over S. Define the functor

$$\widetilde{X} : \mathbf{Sch}_S \to \mathbf{Set}$$

as follows.

- For a scheme Z over S, $\widetilde{X}(Z) = X(Z)$ is the Z-points of X.
- For a morphism of schemes $f : Z \to Y$, $X(f) : X(Z) \to X(Y)$ is given by precomposition with f.

The Yoneda embedding implies the following.

Lemma 6.1.3
Let S be a scheme. Then the Yoneda embedding $\widetilde{(-)} : \mathbf{Sch}_S \to \mathrm{Hom}_{\mathbf{Cat}}(\mathbf{Sch}_S, \mathbf{Set})$ is fully faithful.

In other words, the Z points of a scheme for all Z determine the scheme itself.

We now specialize it to the case where $R = k$ the base field.

Example 6.1.4
The $\mathbb{C}$-points of Spec$(\mathbb{C}[x])$ are in bijection with $\mathbb{C}$.

Proof
The $\mathbb{C}$-points of the affine scheme are in bijection with the ring homomorphisms $\mathbb{C}[x] \to \mathbb{C}$. Clearly for any $a \in \mathbb{C}$, the evaluation homomorphism $\varphi_a : \mathbb{C}[x] \to \mathbb{C}$ is a ring homomorphism. On the other hand, given a ring homomorphism $\phi : \mathbb{C}[x] \to \mathbb{C}$, it is surjective since $\phi(1) = 1$ by definition. Then there is a ring isomorphism $\frac{\mathbb{C}[x]}{\ker(\phi)} \cong \mathbb{C}$ given by the first isomorphism theorem. Since $\mathbb{C}$ is a field, $\ker(\phi)$ is a maximal ideal and by the correspondence of maximal ideals of $\mathbb{C}[x]$, we must have $\ker(\phi) = (x - a)$ for some $a \in \mathbb{C}$. Hence $\phi = \varphi_a$. Hence we are done. ∎

Notice that Spec$(\mathbb{C}[x])$ has an extra point (0) which is not a $\mathbb{C}$-point of the affine scheme.

Example 6.1.5
The $\mathbb{Q}$-points of Spec $\left(\frac{\mathbb{Z}[x,y]}{(x^2+y^2-16)}\right)$ are precisely the rational solutions of $x^2+y^2=16$.

Proof
Indeed, by the fully faithfulness of spec, we have

$$\mathrm{Hom}_{\mathbf{Sch}}\left(\mathrm{Spec}(\mathbb{Q}), \mathrm{Spec}\left(\frac{\mathbb{Z}[x,y]}{(x^2+y^2-16)}\right)\right) \cong \mathrm{Hom}_{\mathbf{CRing}}\left(\frac{\mathbb{Z}[x,y]}{(x^2+y^2-16)}, \mathbb{Q}\right)$$

Suppose that (u,v) is a rational solution to the equation. Then the morphism $\varphi:\mathbb{Z}[x,y]\to\mathbb{Q}$ defined by $\varphi(f)=f(u,v)$ factors through $\frac{\mathbb{Z}[x,y]}{(x^2+y^2-16)}$ and gives rise to a $\mathbb{Q}$-point. Conversely, given any $\mathbb{Q}$-point $\varphi:\frac{\mathbb{Z}[x,y]}{(x^2+y^2-16)}$, precomposition with the quotient map gives a morphism $\mathbb{Z}[x,y]\to\mathbb{Q}$ which must be of the form of an evaluation homomorphism. ∎

Proposition 6.1.6
Let k be a field. Let X be a scheme. Then there is a natural bijection

$$X(k)\cong\{x\in X \mid \text{the canonical homomorphism } k\to k(x) \text{ is an isomorphism }\}$$

Lemma 6.1.7
Let k be an algebraically closed field. Let X be a variety over k. Then there is a natural bijection

$$X(k)\cong\{x\in X \mid x \text{ is a closed point of } X\}$$

6.2 Generic Points and Function Fields

Definition 6.2.1 (Generic Points)
Let X be a space. Let $Z\subseteq X$ be an irreducible closed subset. Let $p\in Z$. We say that p is a generic point for Z if $Z=\overline{\{p\}}$.

The second part of 1.1.12 shows that every irreducible component of an affine scheme has a unique generic point.

Lemma 6.2.2
Let X be a scheme. Then the following are true.
- If $Z\subseteq X$ is an irreducible closed subset, then Z has a unique generic point.
- There is a bijection
$$X \overset{1:1}{\longleftrightarrow} \{\text{ Irreducible closed subsets of } X\}$$
given by $p\mapsto\overline{\{p\}}$.

Proof Let $\{V_i=\mathrm{Spec}(R_i)\mid i\in J\}$ be an open affine cover of X. Then there exists a subset $\{U_i\mid i\in I\}$ of the open cover that forms an open cover for Z and each $Z\cap U_i$ is non-empty. Since Z is irreducible, $Z\cap U_i$ is irreducible. Since Z is non-empty (irreducible spaces are non-empty by assumption), the second case must occur at least in one i, and by 1.1.12 there exists a unique $P_i\in\mathrm{Spec}(R_i)$ such that $Z\cap U_i=\overline{\{P_i\}}^{U_i}$ where $\overline{(-)}^{U_i}$ refers to the closure taken in U_i.

Now I claim that $\overline{T}^{U_i}=\overline{T}^X\cap U_i$ for any subset $T\subseteq U_i$. To prove this, it suffices to show that $\overline{T}^{U_i}$ is the smallest closed subset in U_i that contains T. So let $V\subseteq U_i$ be a closed subset containing T. By definition of the subspace topology of U_i, there exists a closed subset $W\subseteq X$ such that $V=W\cap U_i$. Then we have

$$\overline{T}^X=\bigcap_{T\subseteq C\subseteq X \text{ closed}} C\subseteq W$$

since $T \subseteq V = W \cap U_i \subseteq W$. Then intersecting with U_i gives

$$\overline{T}^X \cap U_i \subseteq W \cap U_i = V$$

hence $\overline{T}^X \cap U_i$ is the closure of T in U_i.

Going back to the main question, we thus have that

$$Z \cap U_i = \overline{\{P_i\}}^X \cap U_i$$

Now I claim that $P_i \in U_j$ for all $j \in I$. Suppose for a contradiction that $P_i \notin U_j$. Then $P_i \in U_i \setminus U_j$. Now since $U_i \cap U_j$ is open, we have that $U_i \setminus U_j$ is closed in U_i. By definition of closure, we have

$$\overline{\{P_i\}}^X \cap U_i = \overline{\{P_i\}}^{U_i} \subseteq U_i \setminus U_j$$

Then we have

$$Z \cap U_i \cap U_j = \overline{\{P_i\}}^{U_i} \cap U_j = \overline{\{P_i\}}^X \cap U_i \cap U_j = \emptyset$$

Taking complements, we get

$$Z = (Z \setminus Z \cap U_i) \cup (Z \setminus Z \cap U_j)$$

Since Z is irreducible, without loss of generality we have $Z = Z \setminus Z \cap U_i$, or $Z \cap U_i = \emptyset$. This is a contradiction since we assumed that are open cover is such that $Z \cap U_i$ is non-empty.

We now have that $P_i \in U_j$ for all $j \in I$. Let $V \subseteq U_i \cap U_j$ be an affine open set, Then we have

$$\overline{\{P_i\}}^V = \overline{\{P_i\}}^X \cap V = Z \cap U_i \cap U_j \cap V = \overline{\{P_j\}}^X \cap V = \overline{\{P_j\}}^V$$

Since V is affine, the generic point is unique by 1.1.12, hence $P_i = P_j$. Write $P = P_i$.

Now since the U_i's cover Z, we have that

$$Z = \bigcup_{i \in I}(Z \cap U_i) = \bigcup_{i \in I}(\overline{\{P_i\}}^X \cap U_i) = \bigcup_{i \in I}(\overline{\{P\}}^X \cap U_i) = \overline{\{P\}}^X$$

Hence Z has a generic point.

For uniqueness, suppose that $Z = \overline{\{p\}}^X = \overline{\{q\}}^X$. Let U be an affine open subset X. Then $Z \cap U$ is affine and

$$\overline{\{p\}}^U = \overline{\{p\}}^X \cap U = \overline{\{q\}}^X \cap U = \overline{\{q\}}^U$$

Since generic points of affine sets are unique, we must have $p = q$.

For the second part of the statement, note that the map $p \mapsto \overline{\{p\}}$ is surjective by the existence of generic point, and the map is injective by uniqueness of the generic point. ■

Definition 6.2.3 (Function Field of an Integral Scheme)
Let X be an integral scheme. Let ν be its unique generic point. Define the function field of X to be

$$k(X) = \mathcal{O}_{X,\eta}$$

Lemma 6.2.4
Let X be an integral scheme. Then the following are true.

- $k(X)$ is a field.
- If $\operatorname{Spec}(R) \cong U \subseteq X$ is a non-empty affine open subset of X, then there is a natural isomor-

phism of rings $k(X) \cong \text{Frac}(R)$ given by the localization map.

Proof
The first point follows from the second point. Let $\varphi : U \xrightarrow{\cong} \text{Spec}(R)$ be the scheme isomorphism. The ring homomorphism is given as follows. Then by 2.4.10 in Sheaf Theory, we have natural isomorphisms

$$\mathcal{O}_{X,\eta} \cong \left(\mathcal{O}_X|_{\text{Spec}(R)}\right)_\eta \cong \mathcal{O}_{\text{Spec}(R),\varphi(\eta)} \cong R_{\varphi(\eta)}$$

where the last isomorphism follows from 1.3.3. Then there is a natural localization map $\mathcal{O}_{X,\eta} \cong R_{\varphi(\eta)} \to \text{Frac}(R)$. Now since X is integral, $R \cong \mathcal{O}_X(\text{Spec}(R))$ is an integral domain and so must have a unique generic point that is the zero ideal (0). Since $\varphi(\eta)$ is also a generic point of $\text{Spec}(R)$, we must have $\varphi(\eta) = (0)$. Then $R_{\varphi(\eta)} = R_{(0)} = \text{Frac}(R)$ so our given map is an isomorphism. ∎

6.3 The Fibers of a Morphism

Definition 6.3.1 (The Fibers of a Morphism)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Let $y \in Y$ be a point. Define the fiber of f at y to be the pullback

$$X_y = X \times_Y \frac{\mathcal{O}_{Y,y}}{m}$$

where m is the unique maximal ideal of $\mathcal{O}_{Y,y}$.

Proposition 6.3.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Let $y \in Y$ be a point. Then there is a natural bijection of sets

$$X_y \cong f^{-1}(y)$$

7 Generic Properties of Morphisms

7.1 Closure Properties of Morphism Classes

Definition 7.1.1 (Stable Under Composition)
Let P be a property of morphisms of schemes. We say that P is stable under composition if the following are true. Suppose that the following is a diagram in **Sch**:

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

If f and g has property P, then $g \circ f$ has property P.

Definition 7.1.2 (Stable Under Base Change)
Let P be a property of morphisms of schemes. We say that P is stable under composition if the following are true. Suppose that the following is a diagram in **Sch**:

$$\begin{tikzcd} X \times_Z Y \arrow[r, "k"] \arrow[d, "h"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

If f has property P, then k has property P.

Definition 7.1.3 (Stable Under Products)
Let P be a property of morphisms of schemes. We say that P is stable under composition if the following are true. Let S be a scheme. Let X_1, X_2, Y_1, Y_2 be schemes. Let $f_1 : X_1 \to Y_1$ and $f_2 : X_2 \to Y_2$ be morphisms. If f_1, f_2 has property P, then

$$f_1 \times_S f_2 : X_1 \times_S X_2 \to Y_1 \times_S Y_2$$

has property P.

Lemma 7.1.4
Let P be a property of morphisms of schemes. Suppose that P is stable under composition and base change. Then P is stable under products.

Proof
Consider the following diagram:

$$\begin{tikzcd} & X \times_S X' \arrow[rr] \arrow[dl] & & X \times_S Y' \arrow[rr] \arrow[dl] \arrow[dr] & & Y \times_S Y' \arrow[dr] & \\ X' \arrow[rr] & & Y' & & X \arrow[rr] & & Y \end{tikzcd}$$

I claim that the two squares are pullback squares. Indeed, there are canonical isomorphisms $X' \times_{Y'} X \times_S Y' \cong X \times_S X'$ and $X \times_Y Y \times_S Y' \cong X \times_S Y'$. Then since the property P is stable under composition and base change, we are done. ∎

Definition 7.1.5 (Local on the Target)
Let P be a property of morphisms of schemes. We say that P is local on the target if the following are true.

- If $f : X \to Y$ has property P, then for all open subsets $V \subseteq Y$, the morphism $f^{-1}(V) \to V$ has property P.
- If $f : X \to Y$ is a morphism and there exists an open cover $Y = \bigcup_{i \in I} V_i$ such that $f^{-1}(V_i) \to V_i$ has property P, then f has property P.

7.2 Affine and Projective Morphisms

Definition 7.2.1 (Affine Morphisms)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is affine if for all affine open subsets $V \subseteq Y$, we have $f^{-1}(V)$ is affine open.

Lemma 7.2.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then f is an affine morphism if and only if there exists an affine open cover $Y = \bigcup_{i \in I} V_i$ such that $f^{-1}(V_i)$ is affine open.

Lemma 7.2.3
The class of affine morphisms satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

7.3 Open, Closed and Locally Closed Embeddings

Definition 7.3.1 (Open Embeddings (Immersion))
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is an open immersion if f induces an isomorphism $X \cong f(X)$ where $f(X)$ is an open subset of Y.

Lemma 7.3.2
The class of open embeddings satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Proof
Let $f : X \to Y$ and $g : Y \to Z$ be open embeddings. Then we know that $X \cong f(X)$ and $Y \cong g(Y)$. Hence $X \cong g(f(X))$. Moreover, since $f(X)$ is an open subset of Y, the isomorphism $Y \cong g(Y)$ implies that $g(f(X))$ is an open subset of $g(Y)$, and hence an open subset of Z. Hence open embeddings are closed under composition.

Let $f : X \to Z$ be an open embedding. Let $g : Y \to Z$ be a morphism. As a space, the pullback of f and g is precisely $g^{-1}(f(X)) \subseteq Y$, and evidently its structure sheaf is given by $\mathcal{O}_Y|_{g^{-1}(f(X))}$. Evidently, the morphism $(g^{-1}(f(X)), \mathcal{O}_Y|_{g^{-1}(f(X))}) \to Y$ is given by inclusion. It is an open embedding since $g^{-1}(f(X))$ is an open subset of Y since $f(X)$ is an open subset of Z.

Let $f : X \to Y$ be an open embedding. Let $V \subseteq Y$ be open. Since $X \cong f(X)$, we must have that $V \cap f(X)$ is the the image of $f^{-1}(V)$ and is an open subset of Y. Conversely, suppose that $f : X \to Y$ is a morphism and $Y = \bigcup_{i \in I} V_i$ is an open cover of Y such that $f^{-1}(V_i) \to V_i$ is an open embedding. Notice that $f^{-1}(V_i)$ for $i \in I$ forms an open cover for X. Let $U \subseteq X$ be open. For each $i \in I$, $U \cap f^{-1}(V_i) \subseteq f^{-1}(V_i)$ is open and its image $f(U \cap f^{-1}(V_i))$ is open. Then $f(U) = \bigcup_{i \in I} f(U \cap f^{-1}(V_i))$ is open. Hence f is a homeomorphism of underlying spaces. Since $\mathcal{O}_X|_U = \mathcal{O}_U$, it follows that the structures sheaves of X and $f(X)$ are also isomorphic. Hence open embeddings are local on target.

Since open embeddings are closed under composition and base change, by the above they must also be closed under products. ■

Definition 7.3.3 (Closed Embeddings (Closed Immersions))
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is a closed embedding if the following are true.
- $X \cong f(X)$ and $f(X) \subseteq Y$ is a closed subset.
- $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective.

Proposition 7.3.4
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then f is a closed embedding if and only if the following are true.
- f is an affine morphism.
- For any affine open subset $V \subseteq Y$, the ring homomorphism $\mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ is surjective.

Lemma 7.3.5
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then f is a closed embedding if and only if there exists an affine open cover $Y = \bigcup_{i \in I} V_i$ such that $f^{-1}(V_i) \to V_i$ is a closed embedding.

Proof
If f is a closed embedding, then $f^{-1}(V_i) \to V_i$ must also be a closed embedding. Conversely, if $f^{-1}(V_i) \to V_i$ is a closed embedding for all $i \in I$, then by 7.2.2 f is an affine morphism. Moreover,
■

Lemma 7.3.6
The class of closed embeddings satisfy the following properties.
- Closed under composition.

Proof
Since surjectivity and affine morphisms are closed under composition, closed embeddings must also be closed under composition.
■

Lemma 7.3.7
Let R, S be commutative rings. Let $\varphi : R \to S$ be a ring homomorphism. If φ is surjective, then Spec(φ) : Spec$(S) \to$ Spec(R) is a closed embedding.

Definition 7.3.8 (Locally Closed Embeddings)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is a locally closed embedding if f factors into a closed embedding followed by an open embedding.

Proposition 7.3.9
Let X, Y be schemes. Let $f : X \to Y$ be a locally closed embedding. If $\text{im}(f)$ is a closed subset of X, then f is a closed embedding.

8 Subschemes of a Scheme

8.1 Open Subschemes

Definition 8.1.1 (Open Subschemes)
Let $(X, \mathcal{O}_X)$ be a scheme. An open subscheme of X is a scheme $(U, \mathcal{O}_X|_U)$ where $U \subseteq X$ is an open subset.

8.2 Closed Subschemes

Given two closed embeddings $f : X \to Y$ and $g : Z \to Y$, even if $f(X) = g(Z)$, the subscheme structure can be different.

Example 8.2.1
Let $f : \text{Spec}(k[x]/(x)) \to \text{Spec}(k[x])$ be induced by the quotient map $k[x] \to k[x]/(x)$. Let $g : \text{Spec}(k[x]/(x^2)) \to \text{Spec}(k[x])$ be the morphism induced by the quotient map $k[x] \to k[x]/(x^2)$. Then the following are true.
- $\text{im}(f) = \text{im}(g)$ as topological spaces.
- $f_*\mathcal{O}_{\text{Spec}(k[x]/(x))}$ and $g_*\mathcal{O}_{\text{Spec}(k[x]/(x^2))}$ are not isomorphic as sheaves.

Proof
Clearly, f is the map defined by $f((0)) = (x)$ and g is the map defined by $g((x)) = (x)$. Hence their images are equal. Moreover, the subsheaf structures are different since their global sections are different. ■

Let X be a scheme. Let $Y \subseteq X$ be a subset. If Y is open, then Y naturally inherits the structure of a scheme. If Y is closed, then it is not obvious how the $\mathcal{O}_X$ is inherited by Y. Indeed, by the above example, a closed subset of a scheme can have multiple subscheme structures. Thus the definition of a closed subscheme must have its structure sheaf built in (compare it to the definition of open subschemes).

Definition 8.2.2 (Closed Subschemes)
Let X be a scheme. A closed subscheme of X is a scheme Y together with a closed embedding $f : Y \to X$ such that $Y \subseteq X$.

Definition 8.2.3 (The Ideal Sheaf)
Let X be a scheme. Let $f : Y \to X$ be a subscheme of X. Define the ideal sheaf of Y to be

$$\mathcal{I}_Y = \ker(\mathcal{O}_X \to f_*\mathcal{O}_Y)$$

Proposition 8.2.4
Let X be a scheme. Then the following are true.
- Let $f : Y \to X$ be a closed subscheme. Then $\mathcal{I}_Y$ is a quasi-coherent sheaf of ideals.
- Let $\mathcal{I}$ be a quasi-coherent sheaf of ideals of X. Then there exists a closed subscheme $Y \subseteq X$ such that $\mathcal{I} \cong \mathcal{I}_Y$.

Proposition 8.2.5
Let R be a commutative ring. Then there is a one to one correspondence

$$\{I \subseteq R \mid I \text{ is an ideal of } R\} \overset{1:1}{\longleftrightarrow} \{Y \subseteq \text{Spec}(R) \mid Y \text{ is a closed subscheme of } \text{Spec}(R)\}$$

given by $I \mapsto \text{Spec}(R/I)$.

Proposition 8.2.6

Let S be a commutative graded ring. Then there is a one to one correspondence

$$\{I \subseteq S \mid I \text{ is a homogeneous ideal of } S\} \overset{1:1}{\longleftrightarrow} \{Y \subseteq \text{Spec}(S) \mid Y \text{ is a closed subscheme of Proj}(S)\}$$

given by $I \mapsto \text{Proj}(S/I)$.

8.3 Locally Closed Subschemes

9 Special Types of Morphisms

9.1 Morphisms of Finite Types

Definition 9.1.1 (Morphisms Locally of Finite Type)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is locally of finite type if for every open affine subset V of Y, there exists an affine open cover $f^{-1}(V) = \bigcup_{i\in I} U_{ij}$ such that $\mathcal{O}_X(U_{ij})$ is a finitely generated $\mathcal{O}_X(V)$-algebra for all i.

Proposition 9.1.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism Then the following are equivalent.
- f is locally of finite type.
- There exists an affine open cover $Y = \bigcup_{i\in I} V_i$ such that there exists an affine open cover $f^{-1}(V_i) = \bigcup_{j\in J} U_{ij}$ such that $\mathcal{O}_X(U_{ij})$ is a finitely generated $\mathcal{O}_Y(V_i)$-algebra.
- For every affine open subset $V \subseteq Y$ and every affine open subset $U \subseteq f^{-1}(V)$, we have $\mathcal{O}_X(U)$ is a finitely generated $\mathcal{O}_Y(V)$-algebra.

Lemma 9.1.3
The class of locally of finite type morphisms satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Definition 9.1.4 (Morphisms of Finite Type)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is of finite type if for every affine open subset V of Y, there is a finite affine open cover $f^{-1}(V) = \bigcup_{i=1}^n U_i$ for $f^{-1}(V)$ such that $mO_X(U_i)$ is a finitely generated $\mathcal{O}_Y(V)$-algebra for $1 \leq i \leq n$.

It is easy to see that we have the following results.

- Let k be a field. There is an equivalence of categories between the category of finitely generated k-algebras and the category of schemes of finite type over k given by the spec functor and the global section functor.
- The restricted Yoneda embedding from the category of schemes of finite type over k to the category of functors
$$\text{Hom}_{\mathbf{Cat}}(\mathbf{FGAlg}_k, \mathbf{Set})$$
is fully faithful.

Lemma 9.1.5
The class of finite type morphisms satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Lemma 9.1.6
Let k be a field. Let X be a scheme of finite type over k. Then $X(k)$ is Noetherian.

9.2 Finite Morphisms

Definition 9.2.1 (Finite Morphisms)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is finite if the following are true.

- f is an affine morphism.
- For every affine open subset $V \subseteq Y$, we have $\mathcal{O}_X(f^{-1}(V))$ is a finitely generated $\mathcal{O}_Y(V)$-module.

Lemma 9.2.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then f is finite if and only if the following are true.
- f is an affine morphism.
- There exists an affine open cover $Y = \bigcup_{i \in I} V_i$ of Y such that $\mathcal{O}_X(f^{-1}(V_i))$ is a finitely generated $\mathcal{O}_Y(V_i)$-module for all $i \in I$.

Lemma 9.2.3
Let X, Y be schemes. Let $f : X \to Y$ be a closed embedding. Then f is finite.

Proof
We have seen that f is affine. Moreover, for any affine open subset $V \subseteq Y$, we have that $\mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ is surjective. Then $\mathcal{O}_X(f^{-1}(V))$ is generated by 1 as an $\mathcal{O}_Y(V)$-module. Indeed, for any $f \in \mathcal{O}_X(f^{-1}(V))$, there exists $g \in \mathcal{O}_Y(V)$ such that g is mapped to f. Then by definition of the module structure, we have $f = g \cdot 1$. Hence $\mathcal{O}_X(f^{-1}(V))$ is generated by 1. ■

Lemma 9.2.4
The class of finite morphisms satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Proof
Let $f : X \to Y$ and $g : Y \to Z$ be finite morphisms. We know that $g \circ f$ is affine. Also, let $W \subseteq Z$ be affine open. Then $\mathcal{O}_Y(g^{-1}(W))$ is a finitely generated $\mathcal{O}_Z(W)$-module, and $\mathcal{O}_X(f^{-1}(g^{-1}(W)))$ is a finitely generated $\mathcal{O}_Y(g^{-1}(W))$-module. Hence $\mathcal{O}_X(f^{-1}(g^{-1}(W)))$ is a finitely generated $\mathcal{O}_Z(W)$-module. Hence $g \circ f$ is finite. ■

Proposition 9.2.5
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. If f is finite, then the following are true.
- For all $y \in Y$, $f^{-1}(y)$ is finite.
- f is closed.

Example 9.2.6
Let k be a field. The inclusion $\mathbb{A}_k^2 \setminus \{(x, y)\} \hookrightarrow \mathbb{A}_k^2$ has finite fibers but is a finite morphism.

Proof
Clearly it has finite fibers. However, the inverse image of $\mathbb{A}_k^2$, which is $\mathbb{A}_k^2 \setminus \{(x, y)\}$ is not affine. Hence it is not an affine morphism and is not finite. ■

Example 9.2.7
The inclusion $\mathbb{A}_\mathbb{C}^1 \setminus \{(x)\} \hookrightarrow \mathbb{A}_\mathbb{C}^1$ has finite fibers and is affine, but it is not a finite morphism.

Proof
Clearly it has finite fibers. It is also affine since $\mathbb{A}_\mathbb{C}^1 \setminus \{(x)\}$ is isomorphic to $\text{Spec}(\mathbb{C}[u, v]/(uv - 1))$ via the morphism $\mathbb{A}_\mathbb{C}^1 \setminus \{(x)\} \to \text{Spec}(\mathbb{C}[u, v]/(uv - 1))$ that is adjoint to the identity $\mathbb{C}[x, y]/(uv - 1) \to \Gamma(\mathbb{A}_C^1 \setminus \{(x)\}) \cong \frac{\mathbb{C}[u,v]}{(uv-1)}$.

Now the morphism $\text{Spec}(\mathbb{C}[u, v]/(uv - 1)) \to \mathbb{A}_\mathbb{C}^1$ is given by the ring homomorphism $\mathbb{C}[x] \to$

$\frac{\mathbb{C}[u,v]}{(uv-1)}$ defined by $x \mapsto (u, 1/u)$. Then $\frac{\mathbb{C}[u,v]}{(uv-1)} \cong \mathbb{C}[u, 1/u]$ is not a finitely generated $\mathbb{C}[x]$-module.

Closed Embeddings	$\subset$	Finite Morphisms	$\subset$	Finite Type Morphisms	$\subset$	Locally of Finite Type Morphisms

9.3 Integral Morphisms

Definition 9.3.1 (Integral Morphisms)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is integral if the following are true.

- f is affine.
- For all affine open subsets $V \subseteq Y$, we have $\mathcal{O}_Y(V)$ is integral over $\mathcal{O}_X(f^{-1}(V))$.

Proposition 9.3.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then f is integral if and only if the following are true.

- f is affine.
- There exists an affine open cover $Y = \bigcup_{i\in I} V_i$ such that $\mathcal{O}_Y(V_i)$ is integral over $\mathcal{O}_X(f^{-1}(V_i))$ for all $i \in I$.

Lemma 9.3.3
The class of integral morphisms satisfy the following properties.

- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Proposition 9.3.4
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then f is finite if and only if the following are true.

- f is a morphism of finite type.
- f is integral.

Proof
Recall from Commutative Algebra 1 that a given a morphism of commutative rings $B \to A$, we have A is a finitely generated B-module if and only if A is a finitely generated B-algebra that is integral over B. Thus we are done.

9.4 Separated Morphisms

Separatedness is essentially the analog of the Hausdorff condition for schemes. Recall that a topological space X is Hausdorff if and only if the digonal morphism to $X \times X$ is closed.

Definition 9.4.1 (Diagonal Morphisms) Let $f : X \to Y$ be a morphism of schemes. The diagonal morphism is the unique morphism $\delta : X \to X \times_Y X$ whose composition with both projection maps $p_1, p_2 : X \times_Y X \to X$ is the identity map of X.

Lemma 9.4.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then $\delta : X \to X \times_Y X$ is a locally closed embedding.

Definition 9.4.3 (Separated Morphisms and Schemes) Let $f : X \to Y$ be a morphism of schemes. We say that f is separated (or X is separated over Y) if the diagonal morphism δ is a closed immersion. A scheme X is separated if it is separated over $\operatorname{Spec}(\mathbb{Z})$.

Proposition 9.4.4 Let $f : X \to Y$ be a morphism of schemes. Then f is separated if and only if the image of the diagonal morphism is a closed subset of $X \times_Y X$.

Proposition 9.4.5
Let X, Y be affine schemes. Let $f : X \to Y$ be a morphism. Then f is separated.

Theorem 9.4.6 (Valuative Criterion of Separatedness)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Suppose that X is Noetherian. Then f is proper if and only if for every valuation ring (R, m) and every morphism $\operatorname{Spec}(R/m) \to X$ and $\operatorname{Spec}(R) \to Y$, any morphism $\operatorname{Spec}(R) \to X$ making the following diagram commutes:

$$\begin{tikzcd} \operatorname{Spec}(R/m) \arrow[r] \arrow[d, "\text{proj.}^\flat"'] & X \arrow[d, "f"] \\ \operatorname{Spec}(R) \arrow[r] \arrow[ur, dashed, "\exists!"] & Y \end{tikzcd}$$

is unique.

Lemma 9.4.7
The class of separated morphisms satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Proposition 9.4.8 Let X and Y be Noetherian schemes. Then any open or closed immersions $f : X \to Y$ are separated.

Lemma 9.4.9
Let R be a commutative ring. Let X be a scheme over R. Then X is separated over $\operatorname{Spec}(R)$ if and only if X is separated over $\operatorname{Spec}(\mathbb{Z})$.

Lemma 9.4.10
Let X be scheme. If X is separated, then the intersection of affine open subsets are affine open.

9.5 Proper Morphisms

Definition 9.5.1 (Universally Closed Morphisms)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is universally close if for any scheme X' and any morphism $g : X' \to Y$, the induced morphism $X \times_Y X' \to X'$ is a closed morphism.

Definition 9.5.2 (Proper Morphisms)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is a proper morphism if f is separated, of finite type and is universally closed.

Let X be a scheme over a field k. The condition for the structure morphism $X \to \operatorname{Spec}(k)$ to be proper is to test that it is separated, of finite type and that for any scheme Y over k, the morphism $X \times_k Y \to Y$ is closed.

Theorem 9.5.3 (Valuative Criterion for Properness)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism of finite type. Suppose that X is Noetherian. Then f is proper if and only if for every valuation ring (R, m) and every morphism $\text{Spec}(R/m) \to X$ and $\text{Spec}(R) \to Y$, there exists a unqiue morphism $\text{Spec}(R) \to X$ such that the following diagram commutes:

$$\begin{tikzcd}
\text{Spec}(R/m) \arrow[r] \arrow[d, "\text{proj.}^\flat"'] & X \arrow[d, "f"] \\
\text{Spec}(R) \arrow[r] \arrow[ur, dashed, "\exists!"] & Y
\end{tikzcd}$$

Lemma 9.5.4
The class of proper morphisms satisfy the following properties.
- Closed under composition.
- Closed under base change.
- Local on the Target.
- Closed under products.

Proposition 9.5.5
Let X, Y be schemes. Let $f : X \to Y$ be a finite morphism. Then f is proper.

Proposition 9.5.6
Let X, Y be affine schemes. Let $f : X \to Y$ be a morphism. Then f is proper if and only if f is finite.

$$\begin{tikzcd}
 & & & \text{Locally of Finite Type Morphisms} & \\
 & & \text{Finite Type Morphisms} \arrow[ur, phantom, sloped, "\subset"] \arrow[r, phantom, "\subset"] \arrow[dl, bend right, "+\text{integral}"'] & \text{Proper Morphisms} \arrow[dr, phantom, sloped, "\subset"] & \\
\text{Closed Embeddings} \arrow[r, phantom, "\subset"] & \text{Finite Morphisms} \arrow[ur, phantom, sloped, "\subset"] \arrow[dr, phantom, sloped, "\subset"] & & & \text{Separated Morphisms} \\
 & & \text{Integral Morphisms} \arrow[r, phantom, "\subset"] & \text{Affine Morphisms} \arrow[ur, phantom, sloped, "\subset"] &
\end{tikzcd}$$

10 Relation to Classical Varieties

10.1 Varieties Revisited

Definition 10.1.1 (The Associated Scheme of a Variety)
Let k be a field. Let X be a variety over k. Define the associated scheme of X as follows.

- The topological space is given by $t(X) = \{Z \subseteq X \mid Z \text{ is irreducible and closed }\}$ whose closed subsets are of the form $t(Z)$ for $Z \subseteq X$ an irreducible and closed subset of X.
- Let $\alpha : X \to t(X)$ be the continuous map $\alpha(p) = \overline{\{p\}}$. Then the sheaf on $t(X)$ is the push-forward sheaf $\alpha_*(\mathcal{O}_X)$.

Definition 10.1.2 (The Associated Scheme of a Variety)
Let k be a field. Define the functor

$$t : \mathbf{Var}_k \to \mathbf{Sch}_k$$

as follows.

- Each variety X over k is associated to the scheme $t(X)$.
- For a morphism of varieties $f : X \to Y$, $t(f)$ is the map defined by $t(f)(Z) = \overline{f(Z)}$ for $Z \subseteq X$ an irreducible closed subset of X.

Proposition 10.1.3
Let k be an algebraically closed field. Then the following are true.

- The functor $t : \mathbf{Var}_k \to \mathbf{Sch}_k$ is fully faithful.
- The essential image of t consists of the Noetherian, reduced and separated schemes of finite type over k.
- Restricted to the essential image, the inverse of t is given by the functor defined by $X \mapsto (X(k), \mathcal{O}_X|_{X(k)})$.
- X is an affine variety if and only if $t(X)$ is an affine scheme.
- X is a projective variety if and only if $t(X)$ is a projective scheme.

This motivates the following definition.

Definition 10.1.4 (Varieties) Let k be a field. A variety over k is a scheme X such that X is an reduced separated scheme of finite type over k.

Lemma 10.1.5
Let k be a field. Let X be a variety over k. Let $x \in X$. Then the following are equivalent.

- x is a closed point of X.
- The field extension $k \to \frac{\mathcal{O}_{X,x}}{m}$ is finite.
- The field extension $k \to \frac{\mathcal{O}_{X,x}}{m}$ is algebraic.

Definition 10.1.6 (The Category of Varieties) Define the category of varieties Var_k over a field k as follows.

- The objects are varieties X over k
- The morphisms are morphisms of schemes $X \to Y$ over k.
- Composition is given by the composition of morphisms.

In virtue of the above proposition, this new category of varieties is equivalent to the category of classical varieties.

Proposition 10.1.7
Let k be a perfect field. Let X, Y be varieties over X. Then the following are true.

- $X \times Y$ is a variety over k.
- If X, Y are affine, then $X \times Y$ is affine.
- If X, Y are projective, then $X \times Y$ is projective.

Proposition 10.1.8
Let k be an algebraically closed field. Let X, Y be varieties over X. Then the following are true.
- If X, Y are irreducible, then $X \times Y$ is irreducible.
- If X, Y are connected, then $X \times Y$ is connected.

Proposition 10.1.9
Let k be a field. Let X, Y be varieties over k. Let $\phi : X \to Y$ be a morphism of schemes over k. Then ϕ is separated and of finite type over k.

Definition 10.1.10 (Subvarities)
Let k be a field. Let X be a variety over k. A subvariety of X is a locally closed subscheme of X that is also a variety.

10.2 Complete Varieties

Definition 10.2.1 (Complete Varieties) Let k be a field. We say that a variety X over k is complete if it is proper over k.

Lemma 10.2.2
Let k be a field. Let X, Y be a varieties over k. If X and Y are complete, then $X \times Y$ is complete.

Proposition 10.2.3
Let k be a field. Let X be a variety over k. Let $Y \subseteq X$ be a subvariety of X. Then the following are true.
- If X is complete and Y is closed in X, then Y is complete.
- If Y is complete, then Y is closed in X.

Lemma 10.2.4
Let k be a field. Let X be an affine variety over k. If X is complete, then $\dim(X) = 0$.

Proposition 10.2.5
Let k be a field. Let X be a variety over k. If X is projective, then X is complete.

11 ???

11.1 Normal Schemes

Definition 11.1.1 (Normal Schemes)
Let X be a scheme. We say that X is normal if for all $p \in X$, $\mathcal{O}_{X,p}$ is a normal domain.

Lemma 11.1.2
Let R be a commutative ring. If R is normal, then $\text{Spec}(R)$ is a normal scheme.

Proof
Let $P \in \text{Spec}(R)$. Since R is normal, the localization R_P is normal. Hence $\mathcal{O}_{X,P} \cong R_P$ is normal. Thus $\text{Spec}(R)$ is a normal scheme. ■

Example 11.1.3
Let k be a field. Let $n \in \mathbb{N} \setminus \{0\}$. Then $\mathbb{A}_k^n, \mathbb{P}_k^n$ are normal schemes.

Proof
Since k is a normal domain, $k[x_1, \dots, x_n]$ is also a normal domain by Commutative Algebra 1. Hence $\mathbb{A}_k^n$ is normal. As for $\mathbb{P}_k^n$, notice that the stalk of any point of $\mathbb{P}_k^n$ can be computed as the stalk of any affine chart $U_i \cong \mathbb{A}_k^n$ containing the point. Since $\mathbb{A}_k^n$ is normal, all stalks are normal domains. Hence all stalks of $\mathbb{P}_k^n$ are normal domains, and $\mathbb{P}_k^n$ is a normal scheme. ■

Example 11.1.4
Let R be a UFD such that $2 \in R$ is invertible. Let $d \in R$ be such that $x^2 - d \in R[x]$ is irreducible. Then $\text{Spec}\left(\frac{R[x]}{(x^2-r)}\right)$ is normal if and only if $d \in R$ is squarefree.

Proof
We have seen from Commutative Algebra 1 7.5.8 that $d \in R$ is squarefree implies that $\frac{R[x]}{(x^2-d)}$ is normal. Hence $\text{Spec}\left(\frac{R[x]}{(x^2-r)}\right)$ is normal.

Conversely, suppose that $d \in R$ is not squarefree. If d is a square, then $\frac{R[x]}{(x^2-d)}$ is not an integral domain and so cannot be normal. Otherwise, without loss of generality we only need to consider the ring $\frac{R[x]}{(x^2-dy^2)}$ where d is squarefree and $y \in R$. In this case, $x^2 - dy^2$ is still irreducible, and since $R[x]$ is a UFD, $(x^2 - dy^2)$ is a prime ideal. But I claim that it is not normal. Consider $\frac{x}{y} \in \text{Frac}\left(\frac{R[x]}{(x^2-dy^2)}\right) = \frac{\text{Frac}(R)[x]}{(x^2-dy^2)}$. It is integral over $\frac{R[x]}{(x^2-dy^2)}$ since it is the root of the monic polynomial $p(t) = t^2 - d \in \frac{R[x]}{(x^2-dy^2)}[t]$. Since normality is a local property, there exists a prime ideal P of $\frac{R[x]}{(x^2-dy^2)}$ such that $\mathcal{O}_{\text{Spec}(R[x]/(x^2-dy^2)),P} \cong R_P$ is not normal. Hence $\text{Spec}\left(\frac{R[x]}{(x^2-r)}\right)$ is not normal.
■

Example 11.1.5
Let k be an algebraically closed field such that $\text{char}(k) \neq 2$. Then the following rings are normal.

- $\text{Spec}\left(\frac{\mathbb{Z}[x]}{(x^2-n)}\right)$ where n is squarefree and $n \equiv 2, 3 \pmod 4$.
- $\text{Spec}\left(\mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right]\right)$ where n is squarefree and $n \equiv 1 \pmod 4$.
- $\text{Spec}\left(\frac{k[x_1,\dots,x_n]}{(x_1^2+\cdots+x_m^2)}\right)$ where $n \geq m \geq 3$.
- $\text{Spec}\left(\frac{k[w,x,y,z]}{(wz-xy)}\right)$.

Proof
From Commutative Algebra 1 7.5.9, we have seen that the first two rings are normal. From 7.5.10 we have seen that the last two rings are normal. ■

Definition 11.1.6 (The Normalization of a Scheme)
Let X be an integral scheme. The normalization of X consists of a scheme $\widetilde{X}$ and a morphism $\pi : \widetilde{X} \to X$ that satisfies the following universal property. For every normal integral scheme Z and every dominant morphism $\phi : Z \to X$, there exists a unique morphism $\widetilde{\phi} : Z \to \widetilde{X}$ such that the following diagram commutes:

$$\begin{tikzcd} Z \arrow[r, dashed, "\exists!\widetilde{\phi}"] \arrow[dr, "\phi"'] & \widetilde{X} \arrow[d, "\pi"] \\ & X \end{tikzcd}$$

Proposition 11.1.7
Let X be an integral scheme. Then the normalization of X exists and is unique (up to unique isomorphism), given by the following gluing data:

- For each affine open subset $U \subseteq X$, define $\overline{U} = \text{Spec}(\overline{\mathcal{O}_X(U)})$ (normalization of $\mathcal{O}_X(U)$ in its fraction field).
- For two affine open subsets $U, V \subseteq X$, define $\theta_{U,V} : \overline{U \cap V} \to \overline{U \cap V}$ by the identity map.

12 The Theory of Divisors

12.1 Weil Divisors

Definition 12.1.1 (Weil Divisors)
Let X be a locally Noetherian integral scheme. Define the group of Weil divisors to be the free abelian group

$$\text{Div}(X) = \mathbb{Z}\langle Y \subseteq X \mid Y \text{ is a closed integral subscheme of codimension one} \rangle$$

generated by closed integral subschemes of X of codimension one.

Example 12.1.2
Let $p = [1 : 0] \in]\mathbb{P}^1$. Let $n > 0$. Then we have $\mathcal{O}_X(np)(U_x) \cong (x/y)^{-n} \cdot k[x/y]$ and $\mathcal{O}_X(np)(U_y) \cong k[(y/x)]$.

Definition 12.1.3 (Divisors of Functions)
Let X be a locally Noetherian integral scheme. For each closed integral subscheme Y of codimension one, let η be the generic point of Y. Let $f \in K(X)^*$. Define the divisor of f to be

$$\text{div}(f) = \sum_Y l_{\mathcal{O}_{X,\eta}}\left(\frac{\mathcal{O}_{X,\eta}}{(f)}\right) \cdot Y$$

If $\mathcal{O}_{X,\eta}$ is a DVR, then

$$l_{\mathcal{O}_{X,\eta}}\left(\frac{\mathcal{O}_{X,\eta}}{(f)}\right) = v(f)$$

where v is the valuation of the DVR $\mathcal{O}_{X,\eta}$.

Example 12.1.4
Let $R = \frac{k[x,y,z]}{(xy-z^2)}$. Then $\text{div}(y) = 2 \cdot \mathbb{V}(y, z)$.

Proof
Let $X = \text{Spec}(R)$. Firstly, notice that $y \in \mathcal{O}_X(X)$ is a globally regular function, hence $v_Z(y) \geq 0$ for all closed integral codimension one subschemes Z. I claim that if $v_Z(y) > 0$, then $Z \subseteq \mathbb{V}(y)$.

Let η be the generic point of Z. Let p_η be the corresponding prime ideal. If $v_Z(y) > 0$, then we have $y \in p_\eta R_{p_\eta}$ which is the maximal ideal of the localization R_{p_η}. Since R is an integral domain, we must have $y \in p_\eta$ and so $p_\eta \in \mathbb{V}(y)$. Then $Z = \overline{p_\eta}^Z \subseteq \overline{p_\eta}^X = \mathbb{V}(y)$.

Thus any non-zero contribution of $\text{div}(y)$ must come from $Z \subseteq \mathbb{V}(y)$ where Z has codimension 1 in X. The only possibility is $Z = \mathbb{V}(y) = \mathbb{V}(y, z)$. Notice that x is invertible in $R_{(y,z)}$. Hence we have $\text{div}(y) = \text{div}(z^2/x) = \text{div}(z^2) = 2$ since (z) is the unique maximal ideal of $R_{(y,z)}$. Hence $\text{div}(y) = 2 \cdot \mathbb{V}(y, z)$. ■

Lemma 12.1.5
Let X be a locally Noetherian integral scheme. Then the map $\text{div} : K(X)^* \to \text{Div}(X)$ given by $f \mapsto \text{div}(f)$ is a well defined group homomorphism.

Definition 12.1.6 (Principal Divisors)
Let X be a locally Noetherian integral scheme. We say that a divisor D on X is principal if $D = \text{div}(f)$ for some function f.

Definition 12.1.7 (The Divisor Class Group)
Let X be a locally Noetherian integral scheme. Let $\text{Prin}(X)$ be the subgroup of all principal divisors of X. Define the divisor class group of X as

$$\text{Cl}(X) = \frac{\text{Div}(X)}{\text{Prin}(X)}$$

Two elements of the same coset are said to be linearly equivalent.

Definition 12.1.8 (Degree Homomorphism)
Let X be a locally Noetherian integral scheme. Define the degree homomorphism

$$\deg : \text{Div}(X) \to \mathbb{Z}$$

by $D = \sum_P n_P \cdot P \mapsto \sum_P n_P$. For each divisor D, define the degree of D to be

$$\deg(D) = \sum_P n_P$$

Definition 12.1.9 (Restrictions of Divisors)
Let X be a locally Noetherian integral scheme. Let D be a divisor on X. Let $U \subseteq X$ be an open subset. Define the restriction of D to U to be

$$D|_U = \sum_{Y \subseteq U} v_Y(f) \cdot Y$$

Definition 12.1.10 (Locally Prinicpal Divisors)
Let X be a locally Noetherian integral scheme. Let D be a divisor on X. We say that D is locally principal if there exists an open cover $\{U_i \mid i \in I\}$ of X such that $D|_{U_i}$ is principal on U_i for all $i \in I$.

12.2 Cartier Divisors

Definition 12.2.1 (The Sheaf of Total Quotient Rings) Let $(X, \mathcal{O}_X)$ be a scheme. Define a presheaf $Q_X : \textbf{Open}(X) \to \textbf{Ring}$ as follows.

- Each $U \subseteq X$ open subset is associated to the total quotient ring $Q_X(U) = Q(\mathcal{O}_X(U))$.
- For each inclusion $U \subseteq V$, define $Q_X(V) \to Q_X(U)$ by just the restriction map.

Define the sheaf of total quotient rings to be the associated sheaf of the presheaf K.

Definition 12.2.2 (Invertible Elements of a Sheaf) Let $(X, \mathcal{F})$ be a ringed space. Define the sheaf of invertible elements

$$\mathcal{F}^* : \textbf{Open} \to \textbf{Grp}$$

of $\mathcal{F}$ as follows.

- For $U \subseteq X$ an open set, define $\mathcal{F}^*(U) =$ The Group of Units of $\mathcal{F}(U)$
- For $U \subseteq V$, define $\mathcal{F}^*(V) \to \mathcal{F}^*(U)$ to just be the restriction map.

Lemma 12.2.3
Let $(X, \mathcal{O}_X)$ be a scheme. Then there is an exact sequence of $\mathcal{O}_X$-modules given by

$$0 \longrightarrow \mathcal{O}_X^* \longrightarrow Q_X^* \longrightarrow \frac{Q_X^*}{\mathcal{O}_X^*} \longrightarrow 0$$

Definition 12.2.4 (Cartier Divisors)
Let X be a scheme. Define the group of Cartier divisors to be the abelian group

$$\text{CDiv}(X) = \Gamma(X, Q_X^*/\mathcal{O}_X^*)$$

Explicitly, a Cartier divisor consists of an open cover $\{U_u \mid i \in I\}$ and rational functions $f_i \in Q_X(U_i)$ such that for each $i, j \in I$, there exists $s \in \mathcal{O}_X^*(U_i \cap U_j)$ such that $f_i = sf_j$.

Example 12.2.5
The set $\{(U_x, 1/(x/y) - 1), (U_y, 1)\}$ is a Cartier divisor on $\mathbb{P}^1$.

Definition 12.2.6 (Principal Cartier Divisors)
Let $(X, \mathcal{O}_X)$ be a scheme. A Cartier divisor f of X is said to be principal if it is the image of the natural map

$$\Gamma(X, Q_X^*) \to \Gamma(X, Q_X^*/\mathcal{O}_X^*) = \text{CDiv}(X)$$

Lemma 12.2.7
Let X be a scheme. Let D be a principal divisor on X. Then there exists $f \in \mathcal{O}_X(X)$ such that D is given by $\{(X, f)\}$.

Definition 12.2.8 (Effective Cartier Divisor)
Let $(X, \mathcal{O}_X)$ be a scheme. A Cartier divisor f of X is said to be effective if it is the image of the natural map

$$\Gamma(X, \mathcal{O}_X \cap Q_X^*) \to \Gamma(X, Q_X^*/\mathcal{O}_X^*) = \text{CDiv}(X)$$

Lemma 12.2.9
Let X be a scheme. Let D be a Cartier divisor. on X Then D is given by the data $\{(U_i, f_i) \mid i \in I\}$ where $X = \bigcup_{i \in I} U_i$ is an open cover, $f_i \in Q_X(U_i)$ such that for $i, j \in I$, there exists $s \in \mathcal{O}_X^*(U_i \cap U_j)$ such that $f_i = sf_j$.

Definition 12.2.10 (Cartier Data)
Let X be a scheme. Let D be a Cartier divisor on X. Define the Cartier data of D to be $\{(U_i, f_i) \mid i \in I\}$ as above.

Lemma 12.2.11
Let X be a scheme. Then the following are true regarding the Cartier divisors on X.

- Let $\{(U_i, f_i) \mid i \in I\}$ and $\{(V_j, g_j \mid j \in J\}$ be Cartier divisors. Then their sum is given by

$$\{(U_i, f_i) \mid i \in I\} + \{(V_j, g_j \mid j \in J\} = \{(U_i \cap V_j \mid f_i g_j) \mid i \in I, j \in J\}$$

- Let D be a Cartier divisor. Then D is principal if and only if $\{(X, f)\}$ is a Cartier data for D.
- Let D be a Cartier divisor. Then D is effective if and only if $\{(U_i, f_i) \mid i \in I\}$ is a Cartier data for D such that $f_i \in \mathcal{O}(U_i)$.

Definition 12.2.12 (The Cartier Divisor Class Group)
Let $(X, \mathcal{O}_X)$ be a scheme. Define the Cartier divisor class group to be the abelian group

$$\text{CCl}(X) = \frac{\text{CDiv}(X)}{\sim}$$

where $D_1 \sim D_2$ if and only if $D_1 - D_2$ is a principal divisor.

12.3 Relation to Subsheaves of the Sheaf of Rational Functions

Definition 12.3.1 (The Associated Sheaf of a Weil Divisor)
Let X be a locally Noetherian integral scheme. Let $D \in \text{Div}(X)$ be a divisor of X. Define the associated sheaf $\mathcal{O}_X(D)$ of D to be the subsheaf of Q_X given by

$$\Gamma(U, \mathcal{O}_X(D)) = \{f \in K(X) \mid f = 0 \text{ or } \text{div}|_U(f) + D|_U \geq 0\} \cup \{0\}$$

for $U \subseteq X$ an open subset.

Lemma 12.3.2
Let X be a locally Noetherian integral scheme. Let D be a divisor on X. Then the following are true.
- $\mathcal{O}_X(D)$ is a quasi-coherent sheaf.
- D is principal if and only if $\mathcal{O}_X(D) \cong \mathcal{O}_X$.
- D is locally principal if and only if $\mathcal{O}_X(D)$ is invertible (a line bundle).

Lemma 12.3.3
Let X be a locally Noetherian integral scheme. Let D, D' be divisors on X. Then D and D' are linearly equivalent if and only if $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$.

Definition 12.3.4 (The Associated Sheaf of a Cartier Divisor)
Let $(X, \mathcal{O}_X)$ be a scheme. Let $D = \{(U_i, f_i \in K_X^*) \mid i \in I\}$ be a Cartier divisor on X. Define the associated sheaf $\mathcal{L}(D)$ of D to be the $\mathcal{O}_X$-module by gluing the schemes $(U_i, \mathcal{O}_X(U_i)[f_i^{-1}])$ for each i.

Proposition 12.3.5
Let X be a scheme. Let D be a Cartier divisor of X. Then $\mathcal{L}(D)$ is an invertible sheaf.

The proposition motivates us to investigate the relations between Cartier divisors and invertible sheaves on a scheme X. This leads to a very fruitful and satisfying result.

Theorem 12.3.6 Let X be a scheme. For any Cartier divisor D of X, the association $D \mapsto \mathcal{L}(D)$ gives a bijection

$$\{\text{Cartier Divisors on } X\} \overset{1:1}{\longleftrightarrow} \{\text{Invertible Subsheaves of } Q_X\}$$

Proposition 12.3.7 Let X be a scheme. Let D_1 and D_2 be Cartier divisors of X. Then there is an isomorphism

$$\mathcal{L}(D_1 - D_2) \cong \mathcal{L}(D_1) \otimes \mathcal{L}(D_2)^{-1}$$

Definition 12.3.8 (The Map from Cartier Divisors to Weil Divisors)
Let X be a locally Noetherian integral scheme. Define the function $d : \text{CDiv}(X) \to \text{Div}(X)$ by

$$\{(U_i, f_i) \mid i \in I\} \mapsto \text{The unique divisor } D \text{ such that } D|_{U_i} = \text{div}|_{U_i}(f_i)$$

Proposition 12.3.9
Let X be a locally Noetherian integral scheme. Then $d : \text{CDiv}(X) \to \text{Div}(X)$ is an injective group homomorphism.

Thus we have inclusions

$$\text{CDiv}(X) \hookrightarrow \substack{\text{Locally Principal} \\ \text{Divisors of } X} \hookrightarrow \text{Div}(X)$$

Proposition 12.3.10
Let X be a normal Noetherian integral scheme. Let D be a Weil divisor on X. Then $D \in \text{im}(d)$ if and only if D is locally principal. In this case, we have $\mathcal{O}(D) = \mathcal{L}(D)$.

12.4 The Divisor Class Groups and the Picard Group

Proposition 12.4.1 Let X be an integral, separated Noetherian scheme such that all local rings are UFD. Then the following are true.

- There is an isomorphism
$$\text{Div}(X) \cong \text{CDiv}(X)$$
induced by d.
- There is an isomorphism
$$\text{Cl}(X) \cong \text{CCl}(X)$$
induced by d.

Proposition 12.4.2
Let X be a scheme. Then the map
$$\text{CCl}(X) \to \text{Pic}(X)$$
defined by $[D] \mapsto \mathcal{L}(D)$ is an injective group homomorphism.

When X is integral, Cartier divisors and the Picard group is entirely the same invariant for X.

Proposition 12.4.3
Let X be an integral scheme. Then the there is an isomorphism of groups
$$\text{CCl}(X) \cong \text{Pic}(X)$$
defined by $[D] \mapsto \mathcal{L}(D)$.

Corollary 12.4.4 Let X be a scheme. If X is Noetherian, integral, separated and that all local rings are UFDs, then there are natural isomorphisms
$$\text{Cl}(X) \cong \text{CCl}(X) \cong \text{Pic}(X)$$

Corollary 12.4.5 Let $X = \mathbb{P}_k^n$ be the projective space over some field k. Then every invertible sheaf on X is isomorphic to $\mathcal{O}_X(m)$ for some $m \in \mathbb{Z}$.

13 Classifying Maps to Projective Space

13.1 Base Points

Definition 13.1.1 (Base Points)
Let X be a scheme. Let $\mathcal{L}$ be an invertible sheaf. Let $p \in X$. We say that p is a base point for $\mathcal{L}$ if $s(p) = 0$ for all $s \in \Gamma(X, \mathcal{L})$.

Definition 13.1.2 (Base Point Free)
Let X be a scheme. Let $\mathcal{L}$ be an invertible sheaf. We say that $\mathcal{L}$ is base point free if $\mathcal{L}$ has no base points.

Lemma 13.1.3
Let X be a scheme. Let $\mathcal{L}$ be an invertible sheaf. Let $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$. Then the following are equivalent.
- $(s_0, \dots, s_n)$ is basepoint free.
- $s_0, \dots, s_n$ has no common zeroes.
- $s_0, \dots, s_n$ generate $\mathcal{L}(X)$.

13.2 Relation to Invertible Sheaves

Proposition 13.2.1
Let R be a commutative ring. Let X be a scheme over R. Then there is a one to one correspondence

$$\{\varphi : X \to \mathbb{P}^n_R \mid \varphi \text{ is an } R\text{-morphism }\} \overset{1:1}{\longleftrightarrow} \frac{\left\{(\mathcal{L}, s_0, \dots, s_n) \;\middle|\; \substack{\mathcal{L} \text{ is an invertible sheaf on } X \\ \text{and } s_0, \dots, s_n \in \mathcal{L}(X) \text{ generate } \mathcal{L}(X)}\right\}}{\sim}$$

given by $\varphi \mapsto (\varphi^*(\mathcal{O}_{\mathbb{P}^n_R}(1)), \varphi^*(s_0), \dots, \varphi^*(s_n))$ and $(\mathcal{L}, s_0, \dots, s_n) \sim (\mathcal{L}', s_0', \dots, s_n')$ if and only if there exists an isomorphism $\mathcal{L} \cong \mathcal{L}'$ such that s_i corresponds to s_i'.

Proposition 13.2.2
Let R be a commutative ring. Let X be a scheme over R. Let $\varphi : X \to \mathbb{P}^n_R$ be an R-morphism corresponding to the invertible sheaf $\mathcal{L}$ of X and global sections $s_0, \dots, s_n$. Then φ is a closed immersion if and only if the following are true.
- For each i, if U_i is the largest open set such that $s_i|_{U_i}$ generates $\mathcal{O}_X|_{U_i}$, then U_i is affine.
- For each i, the map of rings $R[x_0, \dots, x_n] \to \mathcal{O}_X(U_i)$ defined by $x_j \mapsto x_j/x_i$ is surjective.

13.3 Ample and Very Ample Line Bundles

We freely interchange between line bundles and invertible sheaves.

Definition 13.3.1 (Very Ample Sheaves)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Let $\mathcal{L}$ be a sheaf on X. We say that $\mathcal{L}$ is very ample relative to Y if there exists a closed embedding $i : X \to \mathbb{P}^n_Y$ such that $\mathcal{L} \cong i^*\mathcal{O}_Y(1)$

If $Y = \text{Spec}(R)$ is affine, then 12.2.1 shows that $\mathcal{L}$ is very ample if and only if it has global sections that generate $\mathcal{L}$ such that the corresponding morphism is a closed embedding.

Definition 13.3.2 (Ample (Invertible) Sheaves)
Let X be a Noetherian scheme. Let $\mathcal{L}$ be an invertible sheaf. We say that $\mathcal{L}$ is ample if $\mathcal{L}$ is

14 The Study of Smoothness

14.1 The Zariski Tangent Space

Definition 14.1.1 (The Zariski (Co)Tangent Space)
Let X be a scheme. Let $p \in X$. Let m_p be the unique maximal ideal of $\mathcal{O}_{X,p}$.
- Define the Zariski tangent space at p to be
$$T_pX = \left(\frac{m_p}{m_p^2}\right)^*$$
- Define the Zariski cotangent space at p to be
$$T_pX^* = \frac{m_p}{m_p^2}$$

Definition 14.1.2 (The Jacobian Matrix at a Point)
Let k be a field. Let X be a scheme of finite type over k. Let $p \in X$. Let Spec $\left(\frac{k[x_1,\dots,x_n]}{(f_1,\dots,f_r)}\right)$ be an affine neighbourhood containing p. Define the Jacobian matrix of X at p to be the matrix
$$J_pX = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}\Big|_p & \cdots & \frac{\partial f_1}{\partial x_n}\Big|_p \\ \vdots & \ddots & \vdots \\ \frac{\partial f_r}{\partial x_1}\Big|_p & \cdots & \frac{\partial f_r}{\partial x_n}\Big|_p \end{pmatrix}$$

Proposition 14.1.3
Let k be a field. Let X be a scheme of finite type over k. Let $p \in X(k)$ be a k-valued point. Then there is an isomorphism
$$T_pX \cong \text{coker}(J_pX)$$

14.2 Regularity and Smoothness

Definition 14.2.1 (Regular Schemes)
Let X be a scheme. Let $p \in X$. We say that X is regular at p if $\mathcal{O}_{X,p}$ is a regular local ring. We say that X is regular if X is regular at all points $p \in X$.

Proposition 14.2.2
Let k be a field. Let X be a scheme of finite type over k. Let $p \in X(k)$ be a k-valued point. Then the following are equivalent.
- p is a regular point of X.
- $\dim(T_pX) = \dim(X)$.
- If Spec $\left(\frac{k[x_1,\dots,x_n]}{(f_1,\dots,f_r)}\right)$ is an affine open cover containing p, then $\text{rank}(J_pX) = n - \dim(X)$

The Jacobian criterion for checking regularity is only applicable for k-valued points. Motivated by this, we define the notion of a smooth scheme to be the schemes that satisfy the Jacobian criterion at all points.

Definition 14.2.3 (Smooth Schemes)
Let k be a field. Let X be a scheme locally of finite type over k. We say that X is smooth if for all $p \in X$, the affine open subscheme Spec $\left(\frac{k[x_1,\dots,x_n]}{(f_1,\dots,f_r)}\right)$ containing p is such that $\text{rank}(J_pX) = n - \dim(X)$.

The main difference between smooth schemes and regular schemes is that the Jacobian rank criterion is satisfied for all (closed) points for smooth schemes, while the same criterion is only necessary on k-valued points for regular schemes.

Lemma 14.2.4
Let k be a field. Let X be a scheme of finite type over k. Then the following are true.
- If X is smooth, then X is regular.
- If k is algebraically closed, then X is smooth if and only if X is regular.

14.3 The Sheaf of Relative Differentials

Lemma 14.3.1
Let X be a scheme over a scheme S. Let $\Delta : X \to X \times_S X$ be the diagonal morphism. Then there exists an open subset $\Delta(X) \subseteq U \subseteq X \times_S X$ such that $\Delta : X \to U$ is a closed immersion.

Definition 14.3.2 (The Sheaf of Relative Differentials)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Let $U \subseteq X \times_Y X$ be the largest open subset such that $\Delta : X \to U$ is a closed immersion. Let $\mathcal{I}$ be the ideal sheaf of $\Delta : X \to U$. Define the sheaf of relative differentials of f to be

$$\Omega^1_{X/Y} = \Delta^*(\mathcal{I}/\mathcal{I}^2)$$

If f is separated, then $\Delta : X \to X \times_Y X$ is already a closed immersion and there is no need to take an open subset of $X \times_Y X$ to form the ideal sheaf. In this case, we have $\Omega^1_{X/Y} = \mathcal{I}/\mathcal{I}^2$.

Lemma 14.3.3
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Then $\Omega^1_{X/Y}$ is a quasicoherent $\mathcal{O}_X$-module.

There is an alternative way of defining the sheaf of relative differentials.

Lemma 14.3.4
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Let $V \subseteq Y$ be an affine open subset. Let $U \subseteq f^{-1}(V)$ be an affine open subset of X. Then there is a an isomorphism

$$\Omega^1_{X/Y}|_U \cong \widetilde{\Omega^1_{\Gamma(U)/\Gamma(V)}}$$

Example 14.3.5
Let $X = \mathbb{A}^n_Y$. Then $\Omega^1_{X/Y}$ is a free $\mathcal{O}_X$-module of rank n.

Proposition 14.3.6 (The First Fundamental Exact Sequence) Let $f : X \to Y$ and $g : Y \to Z$ be morphism of schemes. Then there is an exact sequence

$$f^*\Omega^1_{Y/Z} \longrightarrow \Omega^1_{X/Z} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

Proposition 14.3.7 (The Second Fundamental Exact Sequence)
Let X, Y, Z be schemes. Let $f : Y \to Z$ be a morphism. Let $i : X \to Y$ be a closed embedding with ideal sheaf $\mathcal{I}$. Then there is an exact sequence of sheaves on X:

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow i_*\Omega^1_{Y/Z} = \Omega^1_{Y/Z} \otimes_{\mathcal{O}_Y} \mathcal{O}_X \longrightarrow \Omega^1_{X/Z} \longrightarrow 0$$

14.4 Smooth Morphisms

Definition 14.4.1 (Smooth Morphisms)
Let k be a field. Let X, Y be schemes locally of finite type over k. Let $f : X \to Y$ be a morphism over k. We say that f is smooth of relative dimension n if the following are true.

- f is flat.
- If $X' \subseteq X$ and $Y' \subseteq Y$ are irreducible components such that $f(X') \subseteq Y'$, then $\dim(X') = \dim(Y') + n$.
- For all $p \in X$, we have
$$\dim_{k(p)}(\Omega^1_{X/Y} \otimes k(p)) = n$$

Smooth morphisms are a direct generalization of smooth schemes.

Lemma 14.4.2
Let k be a field. Let X be a scheme locally of finite type over k. Then the following are equivalent.

- X is smooth.
- The canonical morphism $X \to \text{Spec}(k)$ is smooth.
- Ω^1_X is locally free.

Notice that if X is irreducible, then the third criterion is equivalent to saying that $\dim_k(\Omega^1_{X/Y}) = n$.

Proposition 14.4.3
Let k be an algebraically closed field. Let X, Y be schemes locally of finite type over k. Let $f : X \to Y$ be a morphism over k. Then the following are equivalent.

- f is smooth of relative dimension $\dim(X) - \dim(Y)$.
- $\Omega^1_{X/Y}$ is locally free of rank $\dim(X) - \dim(Y)$.
- For all closed points $p \in X$, the induced map $T_pX \to T_{f(p)}Y$ is surjective.

14.5 Smooth Varieties

Proposition 14.5.1
Let k be a field. Let X, Y be smooth varieties over k. Let $i : X \to Y$ be a closed embedding. Let $\mathcal{I}$ be the ideal sheaf of the closed embedding. Then the second fundamental exact sequence is exact on the left:
$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow \Omega^1_Y \otimes_{\mathcal{O}_Y} \mathcal{O}_X \longrightarrow \Omega^1_X \longrightarrow 0$$

Corollary 14.5.2
Let k be a field of characteristic 0 or a finite field. Let X be a smooth variety over k. Let $p \in X$ be a closed point. Then the second fundamental exact sequence induces an isomorphism
$$\frac{m_p}{m_p^2} \cong \Omega^1_X \otimes_{\mathcal{O}_X} \frac{\mathcal{O}_{X,p}}{m_p}$$
where m_p is the maximal ideal of $\mathcal{O}_{X,p}$.

Theorem 14.5.3 (Euler Sequence) Let A be a commutative ring. Then there is an exact sequence
$$0 \longrightarrow \Omega^1_{\mathbb{P}^n_A/A} \xrightarrow{f} \mathcal{O}_{\mathbb{P}^n_A}(-1)^{n+1} \xrightarrow{g} \mathcal{O}_X \longrightarrow 0$$
where the morphisms are defined as follows.

- The morphism $f : \Omega^1_{\mathbb{P}^n_A/A} \to \mathcal{O}_{\mathbb{P}^n_A}(-1)^{n+1}$ is defined by
- The morphism $g : \mathcal{O}_{\mathbb{P}^n_A}(-1)^{n+1} \to \mathcal{O}_X$ is defined as follows. Let $e_0, \dots, e_n$ be the $A[x_0, \dots, x_n]$-basis of the graded A-module $A[x_0, \dots, x_n]^{n+1}$ (each has degree 1). Then g is induced by the morphism $A[x_0, \dots, x_n]^{n+1} \to A[x_0, \dots, x_n]$ defined by $e_i \to x_i$.

Proposition 14.5.4
Let k be a field of characteristic 0. Let $X \subseteq \mathbb{P}_k^n$ be a hypersurface of degree $d \in \mathbb{N} \setminus \{0\}$. Let $i : X \to \mathbb{P}_k^n$ be the inclusion. Then there is an exact sequence

$$0 \longrightarrow \mathcal{O}_X(-d) \xrightarrow{f} i^*\Omega^1_{\mathbb{P}_k^n/k} \xrightarrow{g} \Omega^1_{X/k} \longrightarrow 0$$

where

14.6 Higher Differential Forms

Definition 14.6.1 (The Differential)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Define the differential $d : \mathcal{O}_X \to \Omega^1_{X/Y}$ to be the morphism of sheaves of abelian groups given by

Note: They are $f^{-1}\mathcal{O}_Y$-linear, but not $\mathcal{O}_X$-linear.

Lemma 14.6.2
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Let $V \subseteq Y$ be an affine open subset. Let $U \subseteq f^{-1}(V)$ be an affine open subset of X. Then $\Gamma(d|_U) : \mathcal{O}_X(U) \to \Omega^1_{X/Y}(U)$ is given by the universal derivation associated to the module of Kahler differentials $\Omega^1_{\Gamma(U)/\Gamma(V)}$.

Definition 14.6.3 (The Relative de Rham Complex)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. Define the relative de Rham complex $\Omega^\bullet_{X/Y} \in \mathbf{Ch}(\mathbf{Ab}(X))$ of X over Y by the following data.
- The nth term is given by $\Omega^n_{X/Y} = \bigwedge_{i=1}^n \Omega^1_{X/Y}$
- The boundary operator $d^n : \Omega^n_{X/Y} \to \Omega^{n+1}_{X/Y}$ is given by $d^n = \bigwedge_{i=1}^n d$

Definition 14.6.4 (The Canonical Sheaf)
Let k be an algebraically closed field. Let X be a regular scheme over k. Define the canonical sheaf of X to be

$$\mathcal{K}_X = \Omega^{\dim(X)}_{X/k} = \bigwedge_{i=1}^{\dim(X)} \Omega^1_{X/k}$$

15 Cohomology of Schemes

Let X be a space. Let $\mathcal{U}$ be an open cover of X. Let $\mathcal{F}$ be a sheaf on X. Recall that there is a natural map

$$\check{\mathrm{H}}^p(X,\mathcal{U},\mathcal{F}) \to H^p(X,\mathcal{F})$$

where $\check{\mathrm{H}}^p(X,\mathcal{U},\mathcal{F})$ is the Čech cohomology of X (the cohomology of the chain complex $C^\bullet(X,\mathcal{U},\mathcal{F}) \in$ **Ch**$(\mathcal{A})$.

Proposition 15.0.1
Let X be a space. Let $\mathcal{U}$ be an affine open cover of X. Let $\mathcal{F}$ be a sheaf on X. If X is Noetherian and separated, then there are isomorphisms

$$\check{\mathrm{H}}^p(X,\mathcal{U},\mathcal{F}) \cong H^p(X,\mathcal{F})$$

for all $p > 0$ induced by the natural map.

15.1 Cohomology of a Noetherian Affine Scheme

Proposition 15.1.1 Let I be an injective module over a Noetherian ring A. Then the sheaf $\widetilde{I}$ on Spec(A) is flasque.

Theorem 15.1.2 Let A be a Noetherian ring. Then for all quasi-coherent sheaves $\mathcal{F}$ on $X =$ Spec(A),

$$H^i(X,\mathcal{F}) = 0$$

for all $i > 0$.

Note that result is also true if we drop the requirement that A is Noetherian. But the proof is more difficult.

Theorem 15.1.3 Let X be a Noetherian scheme. Then the following are equivalent.
- X is an affine scheme
- $H^i(X,\mathcal{F}) = 0$ for all quasi-coherent sheaves $\mathcal{F}$ and all $i > 0$
- $H^1(X,\mathcal{I}) = 0$ for all coherent sheaves of ideals $\mathcal{I}$

15.2 Cohomology of Projective Space

Theorem 15.2.1 Let A be a Noetherian commutative ring and let $S = A[x_0,\dots,x_r]$. Then the following are true.
- For all $n \in \mathbb{Z}$, we have
$$H^i(\text{Proj}(S),\mathcal{O}_{\text{Proj}(S)}(n)) = 0$$
for $0 < i < r$
- There is an isomorphism
$$H^r(\text{Proj}(S),\mathcal{O}_{\text{Proj}(S)}(-r-1)) \cong A$$
- For each $n \in \mathbb{Z}$, the natural map
$$H^0(\text{Proj}(S),\mathcal{O}_{\text{Proj}(S)}(n)) \times H^r(\text{Proj}(S),\mathcal{O}_{\text{Proj}(S)}(-n-r-1)) \to H^r(\text{Proj}(S),\mathcal{O}_{\text{Proj}(S)}(-r-1)) \cong A$$
is a perfect pairing of finitely generated free A-modules

15.3 The Serre Duality Theorems

Theorem 15.3.1 (Serre Duality for Projective Spaces)
Let k be a field.

15.4 Algebraic de Rham Cohomology

Definition 15.4.1 (Algebraic de Rham Cohomology)
Let S be a scheme. Let X be a scheme over S. Define the nth algebraic de Rham cohomology group by
$$H^n_{\text{dR}}(X/S) = \mathbb{H}^n(X;\Omega^\bullet_{X/S})$$

Lemma 15.4.2
Let S be a scheme. Let X be a scheme over S. Then $\text{Tot}^{\oplus}\left(\mathcal{C}^\bullet(X,\mathcal{U},\Omega^\bullet_{X/S})\right)$ is an acyclic resolution of $\Omega^\bullet_{X/S}$.

Thus one way to compute the de Rham cohomology would be
$$H^n_{\text{dR}}(X/S) \cong H^n(\text{Tot}^{\oplus}(\mathcal{C}^\bullet(X,\mathcal{U},\Omega^\bullet_{X/S}))$$

Example 15.4.3
Let $U_0 = \text{Spec}(R_0 = k[x,y]/(y^2 - f(x)))$ where k is a field of characteristic 0 and f is separable and $\deg(f) = 2g+1$ or $2g+2$. Let $U_1 = \text{Spec}(R_1 = k[u,v]/(v^2 - h(u)))$ where $h(u) = u^{2g+2}h(1/u)$. Then we have
$$H^i_{\text{dR}}(X/k) = H^i\left(R_0 \oplus R_1 \to R_0\frac{dx}{y} \oplus R_1\frac{du}{v} \oplus R_0[1/x] \to R_0[1/x]dx\right)$$
where the first map is given by $(w,z) \mapsto (dw, dz, w - z(1/x, y/x^{g+1}))$, the second map is given by $(wdx, zdu, g(x,y)) \mapsto wdx - zdu - dg$ where $du = -z(1/x, y/x^{g+1})dx/x^2$.

- $H^0_{\text{dR}}(X/k) \cong k$.
- $H^2_{\text{dR}}(X/k) \cong H^1_{\text{dR}}(X/k)$.
- $H^0(X,\Omega^0_{X/k}) \hookrightarrow H^1_{\text{dR}}(X/k)$ given by $\omega \mapsto (\omega|_{U_0}, \omega|_{U_1}, 0)$ is an inclusion since the global sections are constant.
- There is a map $H^1_{\text{dR}}(X/kk) \to H^1(X,\mathcal{O}_X$ given by $(\omega_0,\omega_1,g) \mapsto g$ and precomposing with the above map gives 0. It is clearly surjective.
- So we have an exact sequence
$$0 \to H^0(X,\Omega^\bullet_{X/k}) \to H^1_{\text{dR}}(X/k) \to H^1(X,\mathcal{O}_X) \to 0$$
called the Hodge filtration on $H^1_{\text{dR}}(X/k)$. In characteristic p this is more subtle: not all global differentials are closed.
- What can we say about the map $H^1_{\text{dR}}(X/k) \to H^1_{\text{dR}}(U_0/k)$ given by $(\omega_0,\omega_1,g) \mapsto g$.
- The LES in local cohomology gives
$$0 \to H^1(X(\mathbb{C}),\mathbb{C}) \to H^1(U_0(\mathbb{C}),\mathbb{C}) \to H^2(X(\mathbb{C}),U_0(\mathbb{C}),\mathbb{C}) \to H^2(X(\mathbb{C}),\mathbb{C}) \to 0$$
and $H^2(X(\mathbb{C}),U_0(\mathbb{C}),\mathbb{C}) \cong H^0((X-U_0)(\mathbb{C}),\mathbb{C}) \cong \mathbb{C}^{|X-U_0|(\mathbb{C})}$. If $\deg(f)$ is odd, $|(X-U_0)(\mathbb{C})| = 1$. Then $H^1(X(\mathbb{C}),\mathbb{C}) \to H^1(U_0(\mathbb{C}),\mathbb{C})$ is an isomrophism. If $\deg(f)$ is even, then we have $0 \to H^1(X(\mathbb{C}),\mathbb{C}) \to H^1(U_0(\mathbb{C}),\mathbb{C}) \to \mathbb{C} \to 0$. Same thing happens for de Rham cohomology. Surjectivity of $H^1_{\text{dR}}(X/k) \to H^1_{\text{dR}}(U_0/k)$ iff every class (ω_0) for $\omega_0 \in \Omega^1_{R_0/k}$ is the image of $\ker(R_0dx/y \oplus R_1du/v \oplus R_0[1/x] \to R_0[1/x]dx)$. i.e. $\exists \omega_1 \in R_1du/v$, $g \in R_0[1/x]$ such that $\omega_0 - \omega_1 = dg$. The obstruction to finding such an ω_1 and g would be the residue of a differential. Pick $z \in (X-U_0)(\overline{k})$ a closed point and a parameter t at z. Write ω as $F(t) = \sum_{i=0}^\infty a_it^i \in \overline{k}[[t]]$. Eg any meromorphic can be written as rational function times dt, then, take formal power series. Then $\text{Res}_z(\omega) = a_1$ of $F(t)$. If $\omega_1 \in \Omega^1_{R_1/k}$ then $\text{Res}_z = 0$ (indeed it has no pole). Also $\text{Res}_z(dg) = 0$. Then image of $H^1_{\text{dR}}(X/k)$ is contained in the

kernel of the residue map

$$\text{Res} : H^0_{\text{dR}}(U_0/k) \to \bigoplus_{Z \in (X - U_0(\overline{k}))} \overline{k}$$

Fact: For any open $U \subseteq X$, $\omega \in H^0(U, \Omega_{X/k})$, then

$$\sum_{Z \in (X-U)(\overline{k})} \text{Res}_z(\omega) = 0$$

Proposition 15.4.4
Let S be a scheme. Let X be a scheme over S. Then the following are true.
- $H^n_{\text{dR}}(X/S)$ is a $H^0_{\text{dR}}(X/S)$-module.
- The map $H^i_{\text{dR}}(X/S) \times H^j_{\text{dR}}(X/S) \to H^{i+j}_{\text{dR}}(X/S)$ induced by the wedge of differential forms is a $H^0_{\text{dR}}(X/S)$-linear.

Definition 15.4.5 (The de Rham Cohomology Ring)
Let S be a scheme. Let X be a scheme over S. Define the de Rham cohomology ring of X to be

$$H^*_{\text{dR}}(X/S) = \bigoplus_{i=0}^{\infty} H^i_{\text{dR}}(X/S)$$

together with ring structure given by wedge of differential forms.

Lemma 15.4.6
Let S be a scheme. Let X be a scheme over S. Then $H^*_{\text{dR}}(X/S)$ is a graded commutative $H^0_{\text{dR}}(X/S)$-algebra.

Lemma 15.4.7
Let k be a field of characteristic 0. Let X be a smooth affine variety over k. Then we have

$$H^n_{\text{dR}}(X) = H^n(\Gamma(X, \Omega^\bullet_{X/k}))$$

for all $n \in \mathbb{N}$.

Example 15.4.8
Let k be a field of characteristic 0. Let $R = k[x_1, \dots, x_n]$ be the polynomial ring over k. Then $\Gamma\left(X, \Omega^1_{\text{Spec}(R)/k}\right) = \Omega^1_{R/k} = \bigoplus_{i=1}^n R dx_i$. Then we have

$$\Gamma\left(X, \Omega^j_{\text{Spec}(R)/k}\right) = \bigoplus_{1 \leq i_1 < \dots < i_j \leq n} R dx_{i_1} \wedge \dots \wedge dx_{i_j}$$

Then we have

$$H^i_{\text{dR}}(R/k) = \begin{cases} k & \text{if } i = 0 \\ 0 & \text{otherwise} \end{cases}$$

TBR:

Example 15.4.9
Let k be a field of characteristic $p > 0$. Then we have

$$\dim_k(H^1(\Gamma(\text{Spec}(k[t]), \Omega^1_{\text{Spec}(k[t])/k}))) = \infty$$

Indeed, we have

$$H^1(\Gamma(\text{Spec}(k[t]), \Omega^1_{\text{Spec}(k[t])/k})) = \frac{k[t]}{\langle f'(t) \mid f \in k[t]\rangle} = \frac{k[t]}{\bigoplus_{i=1}^\infty it^{i-1}dt} \cong \bigoplus_{i=1}^\infty t^{ip-1}dt$$

Example 15.4.10
$\Omega^1_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}} \cong (\mathbb{Z}/2\mathbb{Z})d\sqrt{2}$. $H^1(\Gamma(\text{Spec}(\mathbb{Z}\sqrt{2}), \Omega^1_{\mathbb{Z}[\sqrt{2}]/\mathbb{Z}})) \neq 0$

Example 15.4.11
Let k be a field of characteristic $p \neq 2$. Let $f \in k[x]$ be separable. Let $R = \frac{k[x,y]}{y^2 - f(x)}$. Then we have

$$\Omega^1_{R/k} \cong \frac{\Omega^1_{k[x,y]/k} \otimes_{k[x,y]} k}{(2ydy - f'(x)dx)} \cong \frac{Rdx \oplus Rdy}{R(2ydy - f'(x)dx)}$$

Using the quotient relation, we may write $dy = \frac{f'(x)dx}{2y}$ in the module. Thus dx/y is a generator of the module.

Since f is separable, there exists $g, h \in k[x]$ such that $gf + hf' = 1$. Let $\omega = aydx + 2bdy \in \Omega^1_{R/k}$. Then we have $y\omega = dx$. Hence y divides dx and $\omega = dx/y$.

Now let $z = a(x) + b(x)y \in R$ be arbitrary. Then we have

$$dz = (a'(x) + b'(x)y)dx + h(x)dy = (ya'(x) + b'(x)f(x) + \frac{1}{2}b(x)f'(x))\omega$$

Thus every element of $\Omega^1_{R/k}$ is of the form $q(x)\omega$ for some $q(x) \in k[x]$. If $\deg(q) \geq \deg(f)$, then there exists $p \in k[x]$ such that $\deg(p) > \deg(q)$ such that

$$[p(x)\omega] = [q(x)\omega]$$

in $H^1_{\text{dR}}(R/k)$.

Lemma 15.4.12
Let k be a field of characteristic 0. Let X be a smooth variety over k. Then the following are true.

- $H^i_{\text{dR}}(X) = 0$ for all $i > 2\dim(X)$.
- If X is affine, then $H^i_{\text{dR}}(X) = 0$ for all $i > \dim(X)$.

Proposition 15.4.13

15.5 Hodge de Rham Spectral Sequences

Definition 15.5.1 (The Hodge Filtration)
Let S be a scheme. Let X be a scheme over S. Define the Hodge filtration $F^\bullet\Omega^\bullet_{X/S}$ of X to be

$$F^i\Omega^j_{X/S} = \begin{cases} \Omega^j_{X/S} & \text{if } j \geq i \\ 0 & \text{otherwise} \end{cases}$$

Thus for each i, the collection of complexes

$$F^i\Omega^\bullet_{X/S} = (0 \to \Omega^i_{X/S} \to \Omega^{i+1}_{X/S} \to \cdots)$$

form the filtration for $\Omega^\bullet_{X/S}$.

Lemma 15.5.2
Let S be a scheme. Let X be a scheme over S. Then $F^\bullet\Omega^\bullet_{X/S}$ is an exhaustive filtration.

Definition 15.5.3 (The Hodge de Rham Spectral Sequence)
Let S be a scheme. Let X be a scheme over S. Define the Hodge de Rham spectral sequence to be the spectral sequence associated to the Hodge siltration $F^\bullet\Omega^\bullet_{X/S}$.

The E^0 page is given by

$$E_0^{p,q} = \frac{F^p\Omega^{p+q}_{X/S}}{F^{p+1}\Omega^{p+q}_{X/S}} = \frac{\Omega^q_{X/S}}{\Omega^{q-1}_{X/S}}$$

Lemma 15.5.4
Let S be a scheme. Let X be a scheme over S. Then we have

$$E_1^{p,q} = H^q(X, \Omega^p_{X/S}) \Rightarrow H_{\text{dR}}(X/S)$$

Theorem 15.5.5 (Deligne)
If $S = \text{Spec}(k)$, k char 0, $X \to S$ smooth and proper. Then the Hodge de Rham spectral sequence degenerates at the E_1 page (all differentials in page 1 is 0).

When you have a 2-step filtration: $0 \to F^1C^\bullet \to C^\bullet \to C^\bullet/F^1C^\bullet \to 0$, the spectral sequence is the long exact sequence in cohomology.

16 Intersection Theory

16.1 The Divisor of a Rational Function

Definition 16.1.1 (The Order of a Function at a Subvariety)
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f/g \in K(W)^*$ for $f, g \in \mathcal{O}_X(W)$ and $g \neq 0$. Let $V \subseteq W$ be an irreducible subvariety of codimension 1. Define the order of f at V to be

$$\text{ord}_V(f/g) = l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(f)}\right) - l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(g)}\right)$$

Definition 16.1.2 (The Divisor of a Rational Function)
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f \in K(W)^*$. Define the divisor of f to be

$$\text{div}(f) = \sum_{\substack{V \text{ is a codimension 1} \\ \text{irreducible subvariety of } W}} \text{ord}_V(f) \cdot V$$

16.2 The Chow Groups

Definition 16.2.1 (The Group of n-Cycles)
Let k be a field. Let X be a scheme of finite type over k. Let $n \in \mathbb{N}$. Define the group of n-cycles on X to be the free abelian group

$$Z_n(X) = \mathbb{Z}\langle V \mid {}^{V \text{ is an } n\text{-dimensional}}_{\text{irreducible subvariety of } X} \rangle$$

Lemma 16.2.2
Let k be a field. Let X be a scheme of finite type over k. Let W be an irreducible subvariety of X. Let $f \in K(W)^*$. Then $\text{div}(f) \in Z_{\dim(W)-1}(X)$.

Definition 16.2.3 (Cycles Rationally Equivalent to Zero)
Let k be a field. Let X be a scheme of finite type over k. Let $V \in Z_n(X)$. We say that V is rationally equivalent to 0 if there exists finitely many $\dim(V) + 1$-dimensional irreducible subvarieties $W_1, \dots, W_r$ of X and $f_i \in K(W_i)^*$ such that

$$V = \sum_{i=1}^n \text{div}(f_i)$$

Definition 16.2.4 (The Chow Group)
Let k be a field. Let X be a scheme of finite type over k. Define the nth Chow group of X to be

$$\text{CH}_n(X) = \frac{Z_n(X)}{\sim}$$

where we say that $V_1 \sim V_2$ if $V_1 - V_2$ is rationally equivalent to 0. In this case, we say that V_1 and V_2 are rationally equivalent.

16.3 The Ring Structure

16.4 The Pushfoward Map

17 ???

17.1 Projective Morphisms

Definition 17.1.1 (Projective Morphisms)
Let X, Y be schemes. Let $f : X \to Y$ be a morphism. We say that f is projective if there exists $n \in \mathbb{N} \setminus \{0\}$ such that f factors into morphisms

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, dashed, "\exists g"'] & & Y \\ & \mathbb{P}^n_Y \arrow[ur, "\text{proj.}"'] & \end{tikzcd}$$

where g is a closed immersion.